HSLU Lucerne University of Applied Sciences and Arts

Computer Vision & AI

SW 03: Color

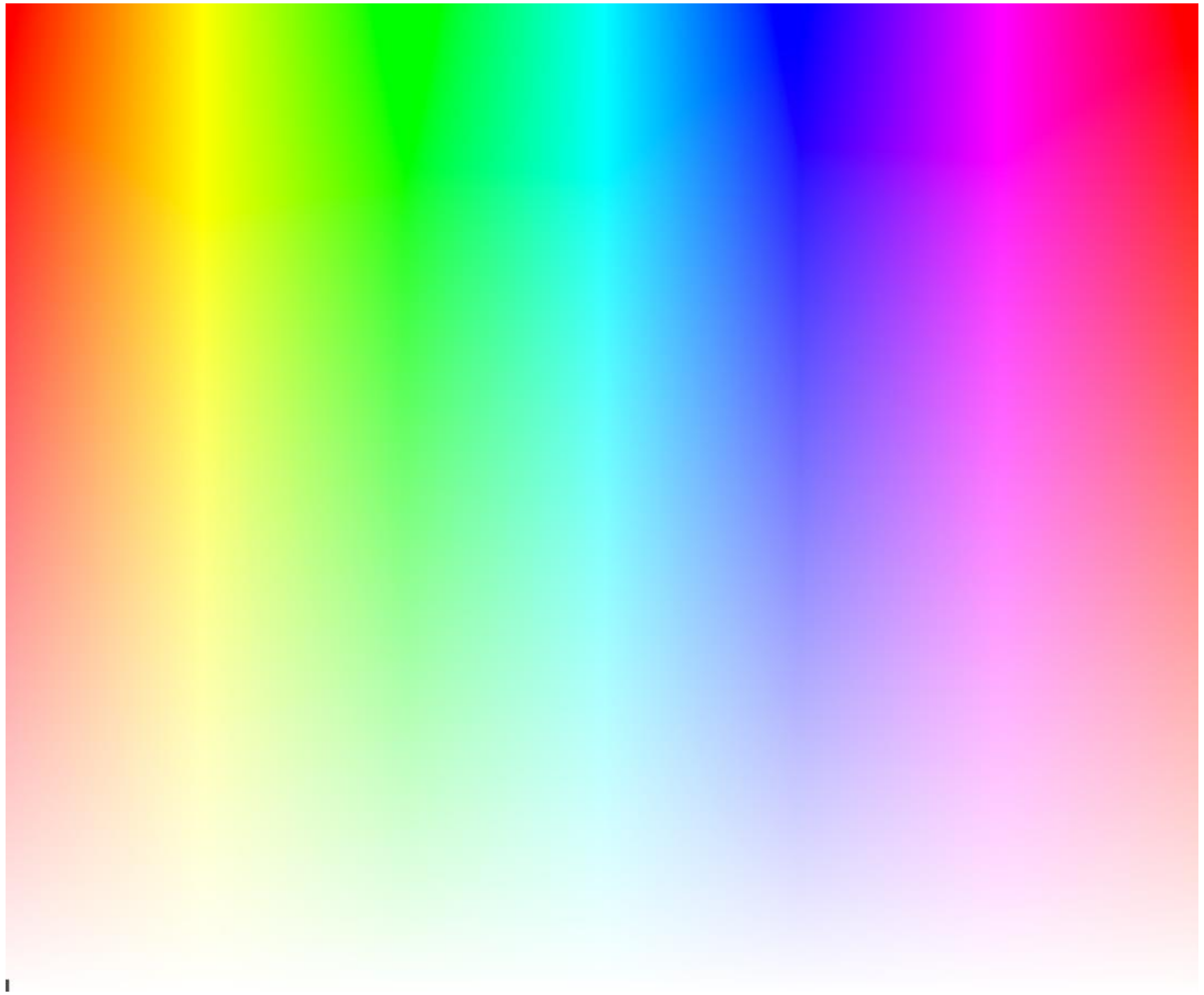
7. März 2024

FH Zentralschweiz

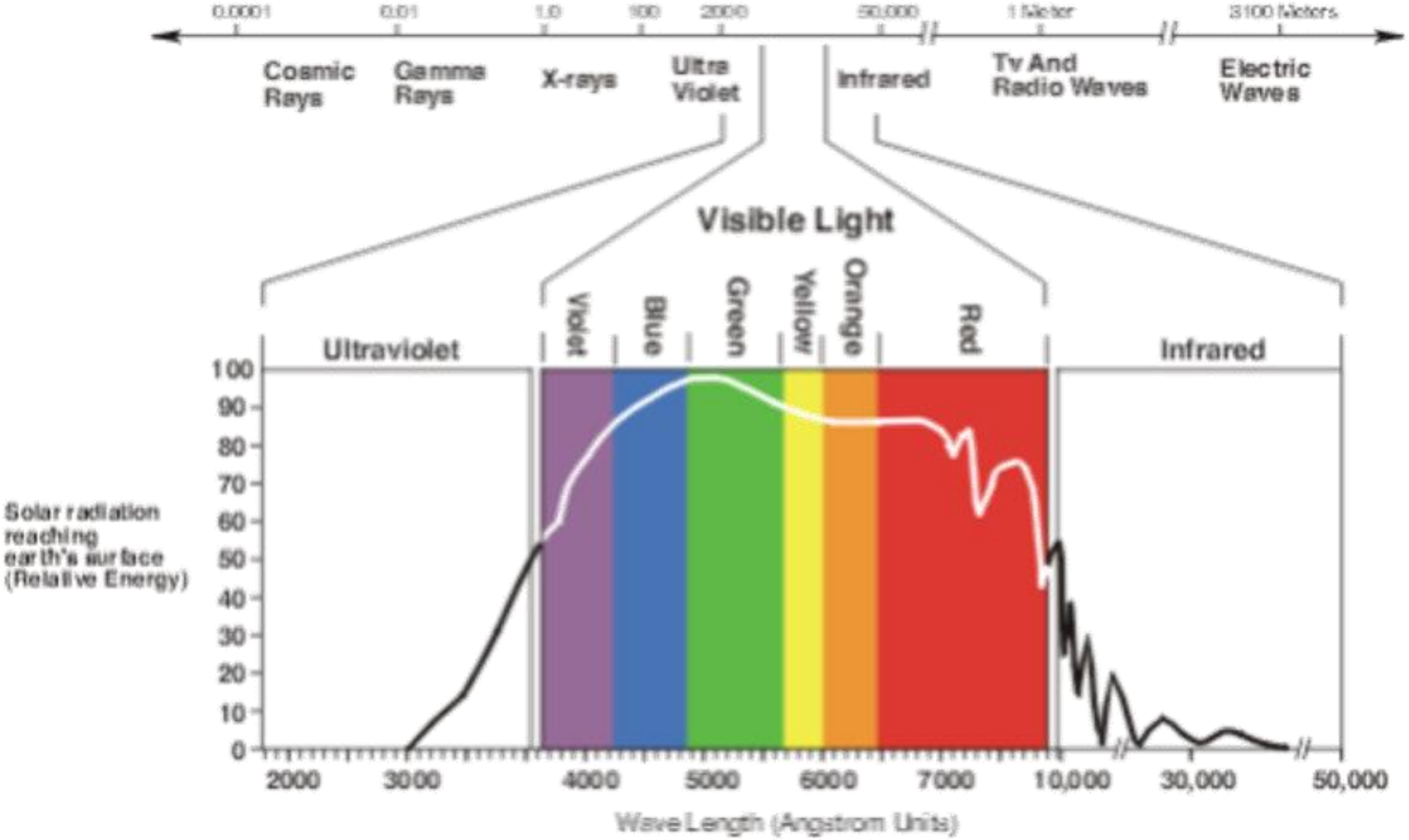
Color

- What is color?

- How do you describe color?



Electromagnetical (Solar) Radiation



Ångström vs Nanometer

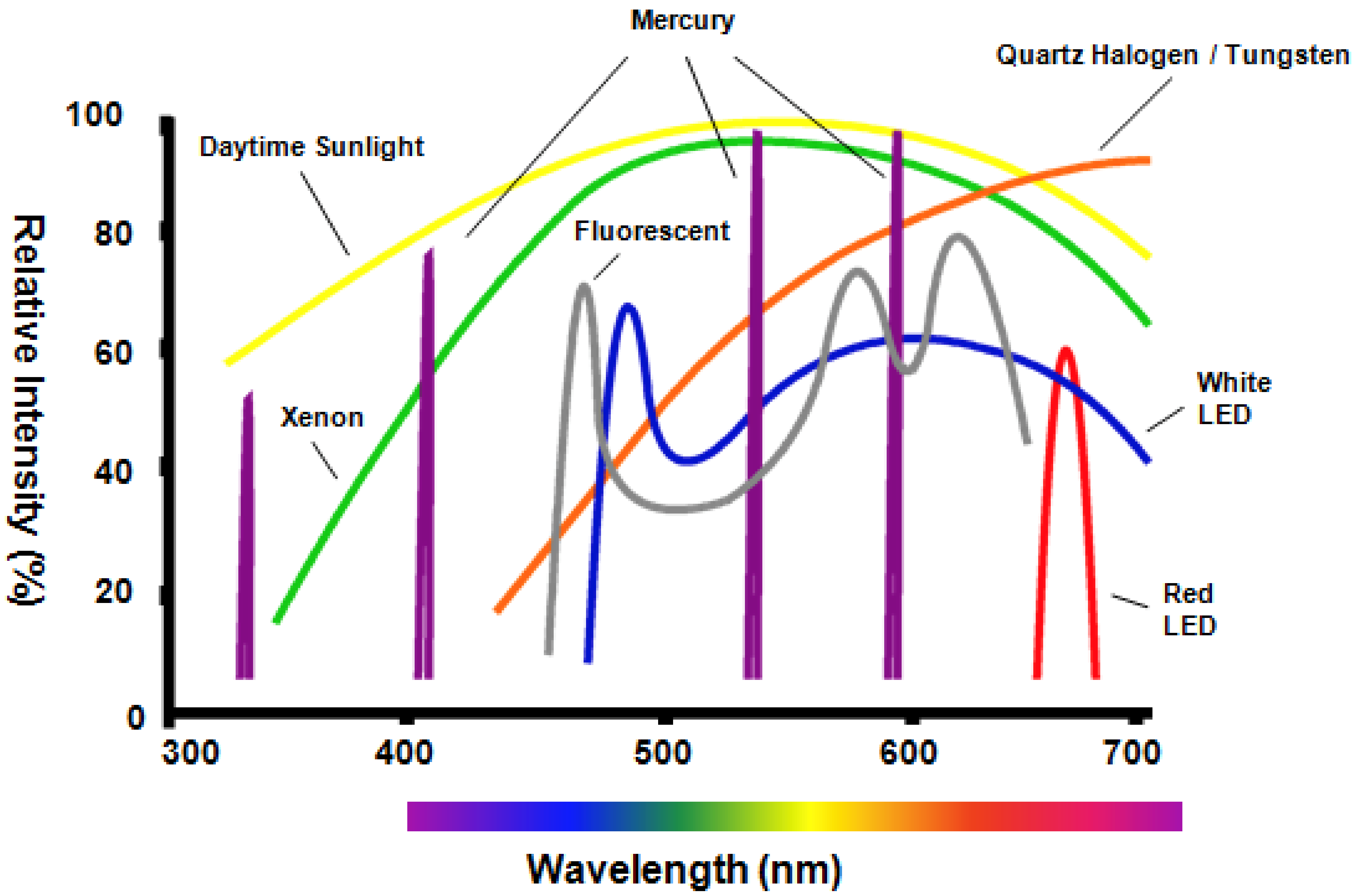
Ångström

- Used in english speaking regions of the world
- not part of the international systems of units
- 1 Å = 10^{-10} m (approximately 1 atomic radius)

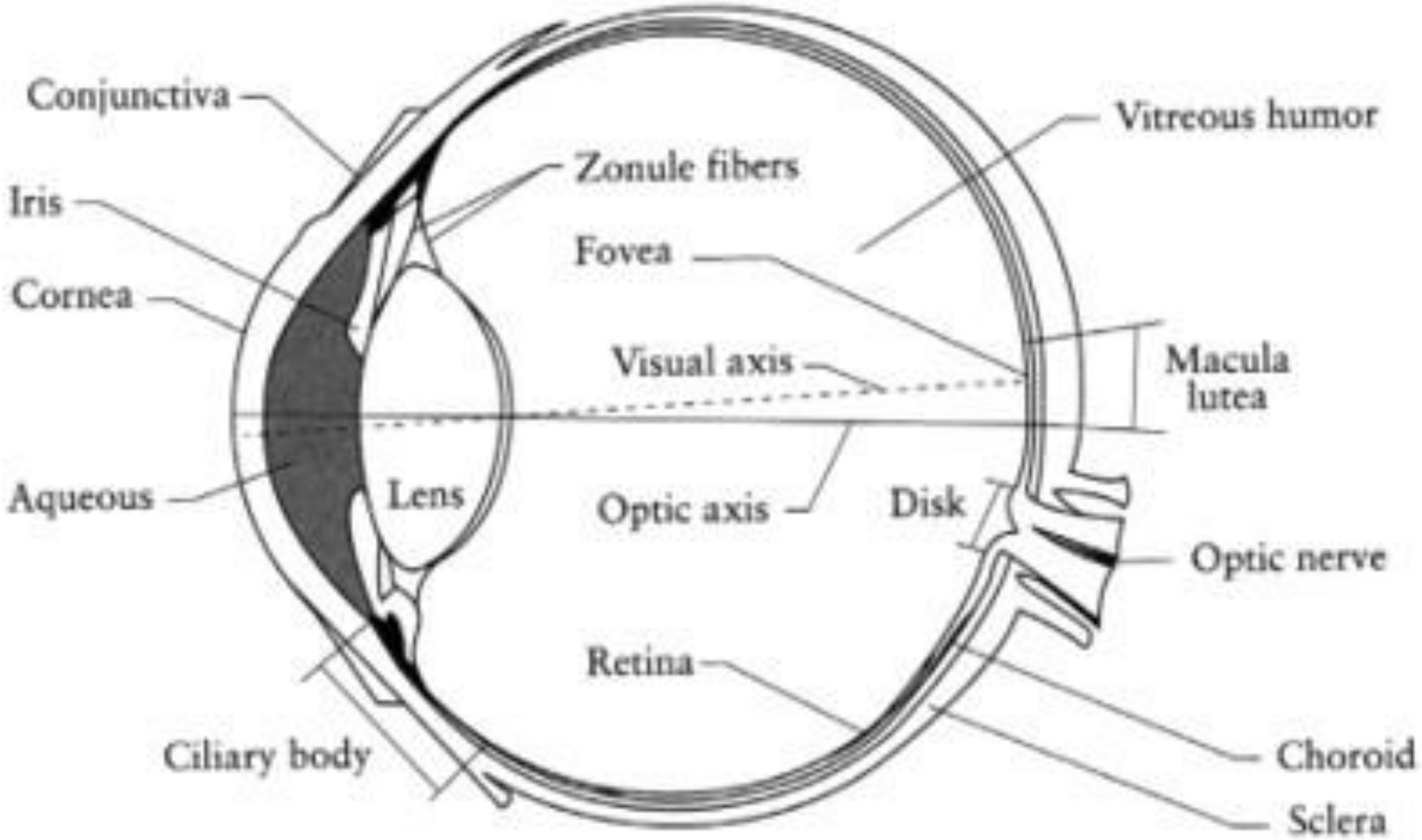
Nanometer

- part of the international systems of units
- 1 nm = 10^{-9} m = 10 Å

Spectral Distribution of Different Light Sources

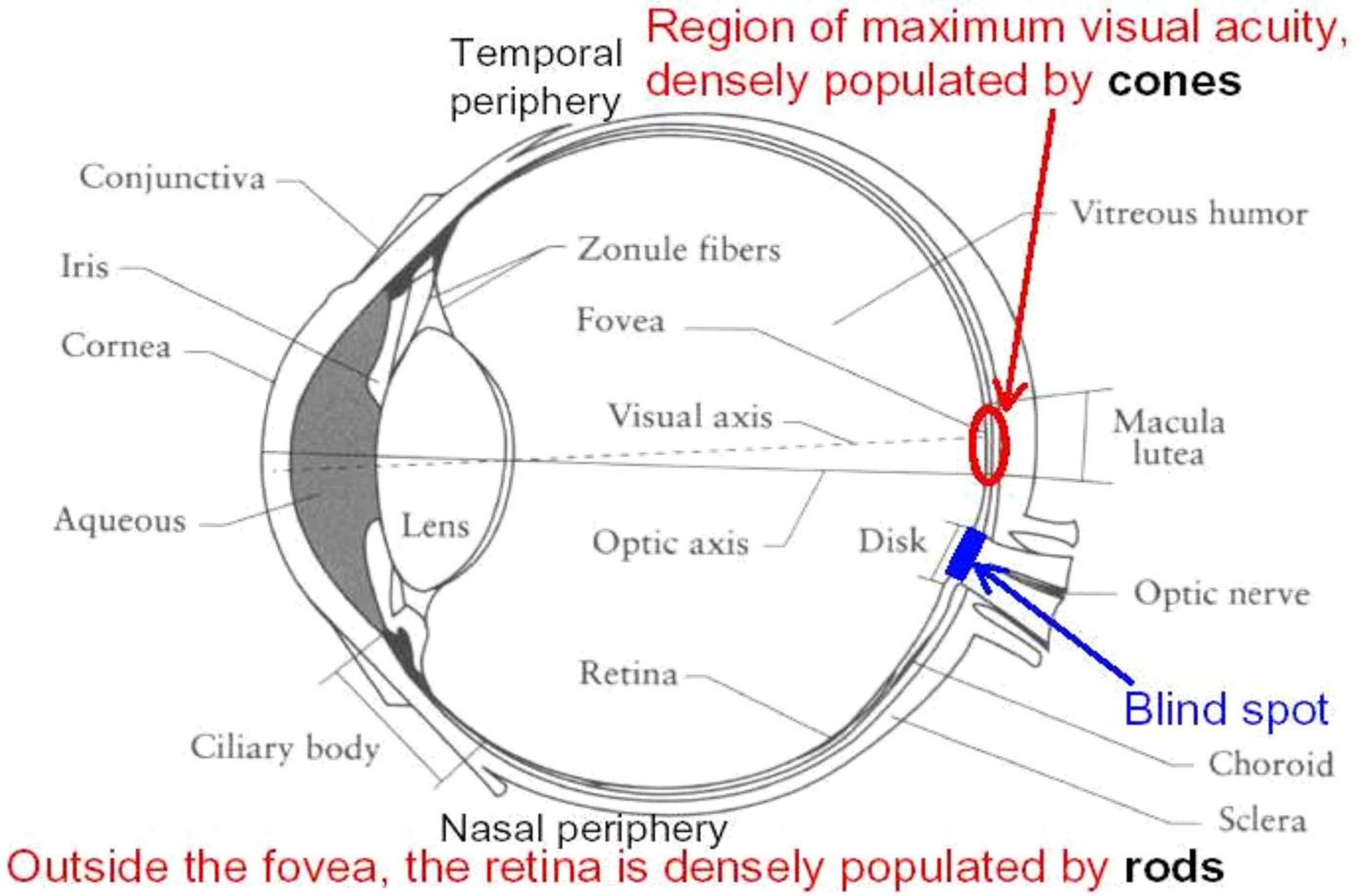


The Human Eye as a Camera

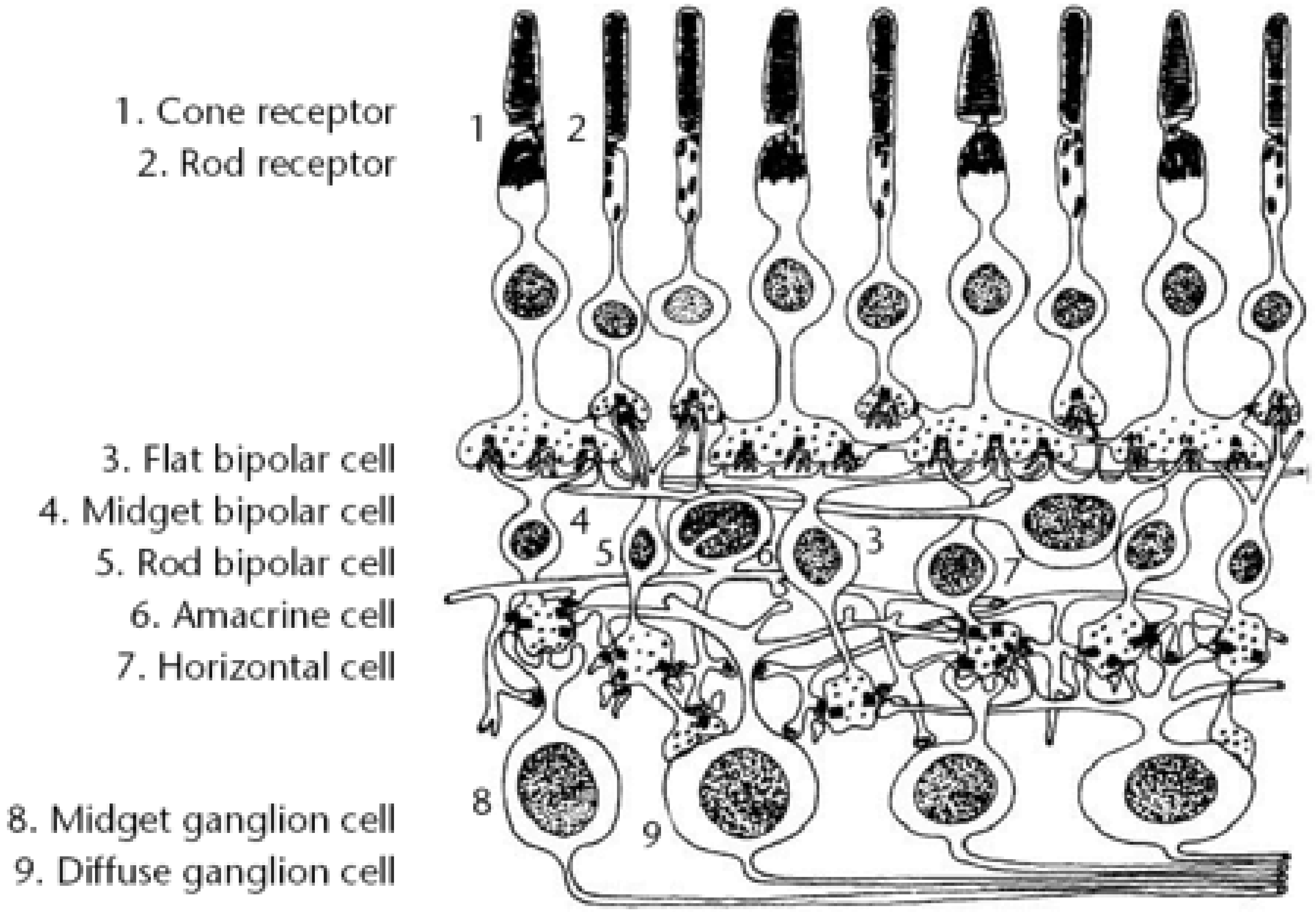


- Iris
  Layer of tissue connected with radial muscles: sphincter and dilator
- Pupil
  Aperture, controlled by iris
- Lens
  Deformable to change focus
- Retina
  contains photoreceptor cells for perception of light

Color Perception (1)

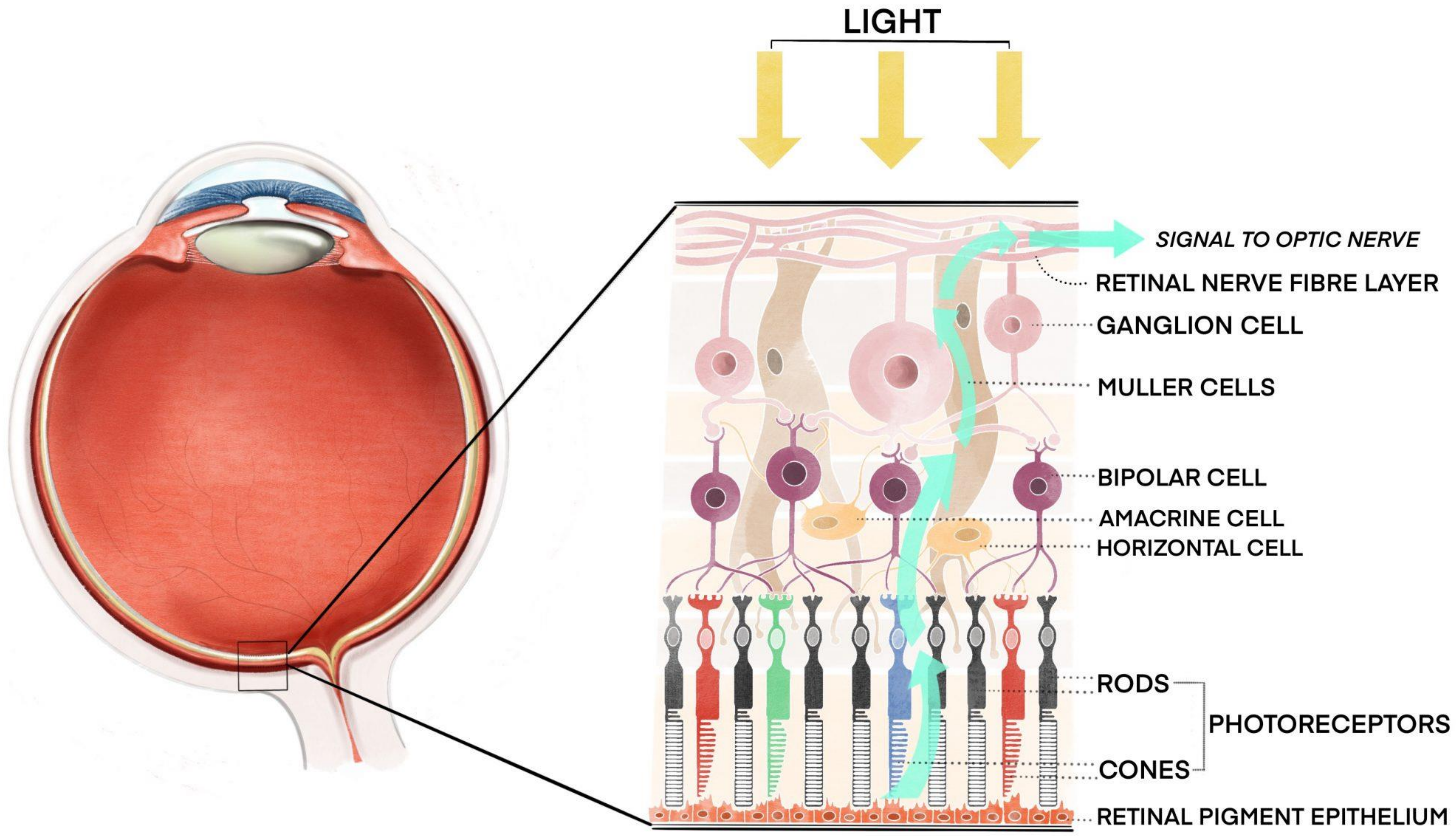


Cross-sectional view of the human eye

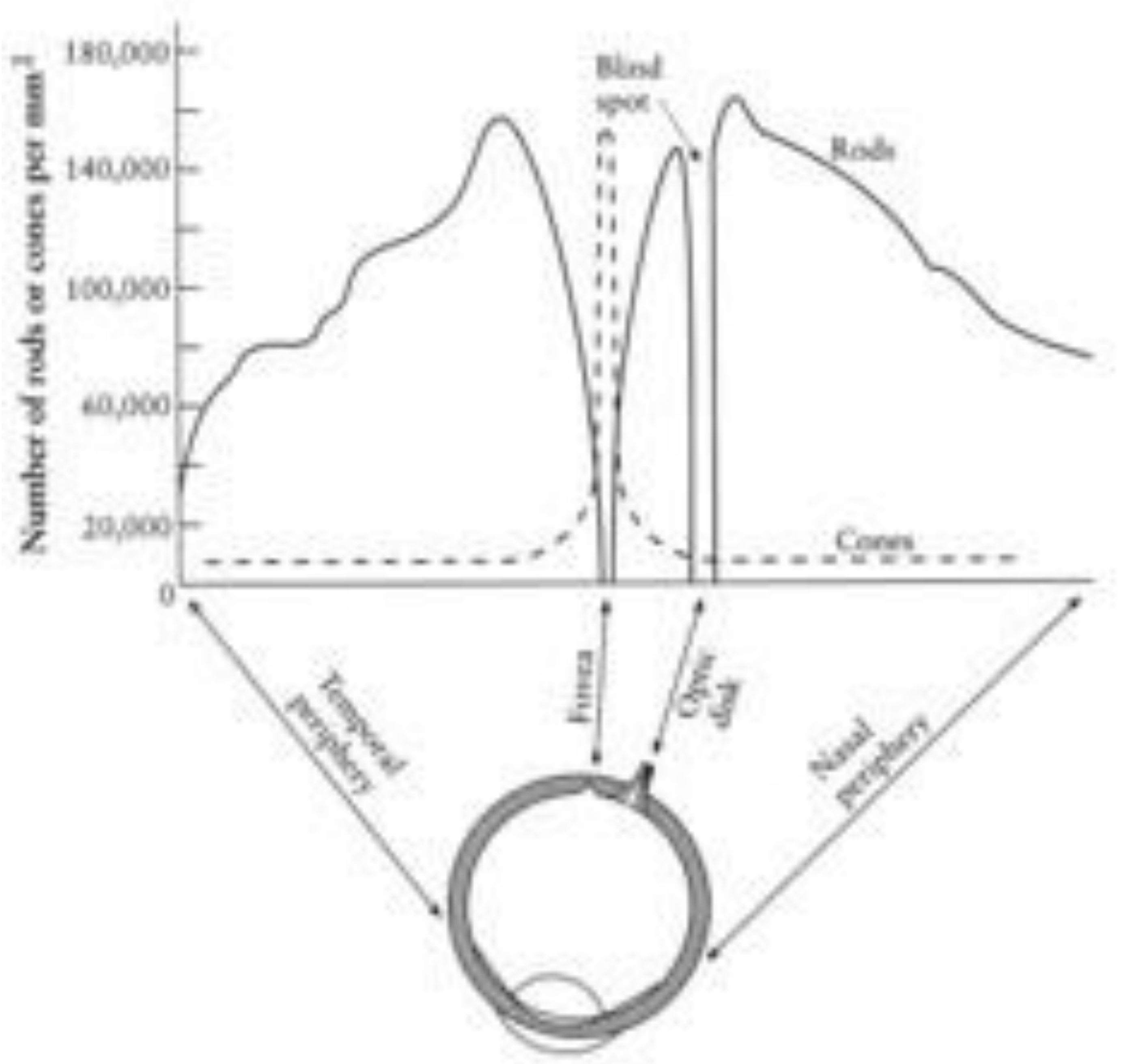


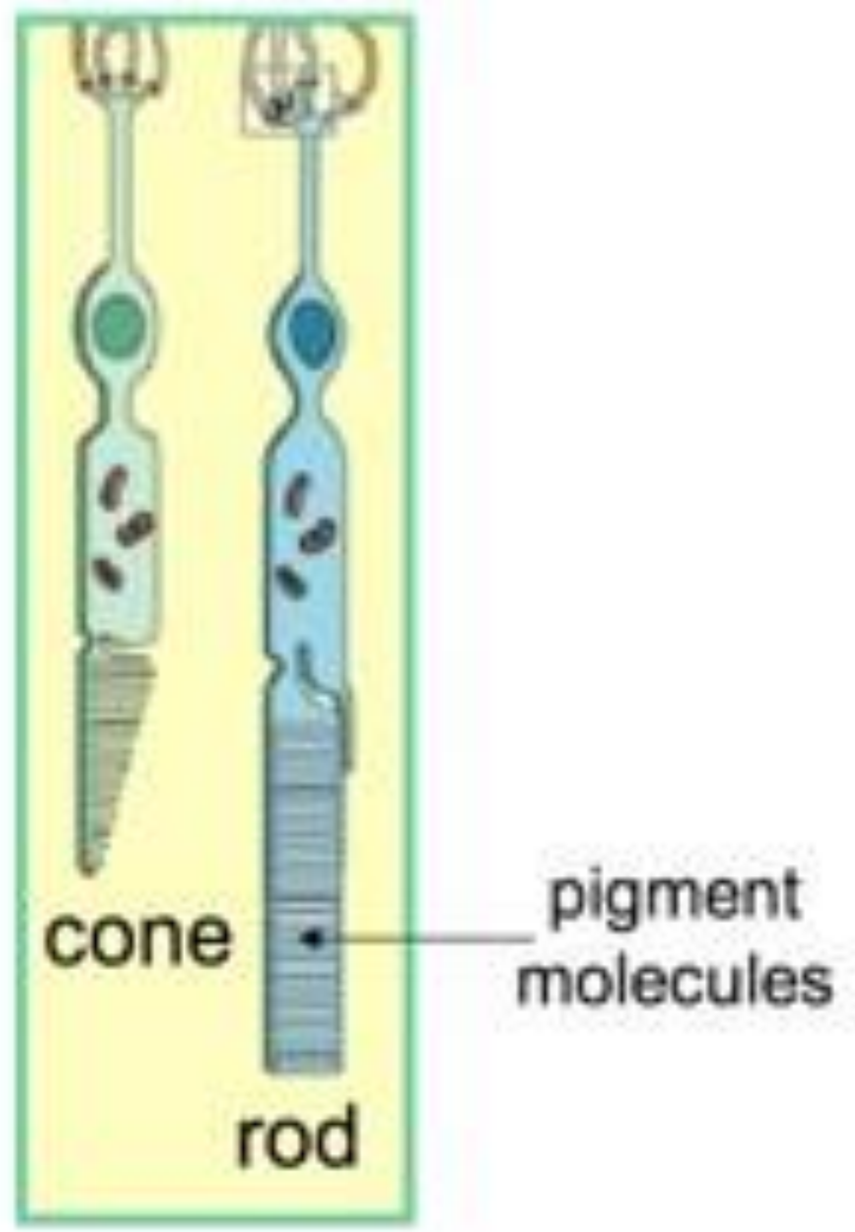
Structure of retina:
rods and cones perform various tasks
in perception of color and light.

Color Perception (2)



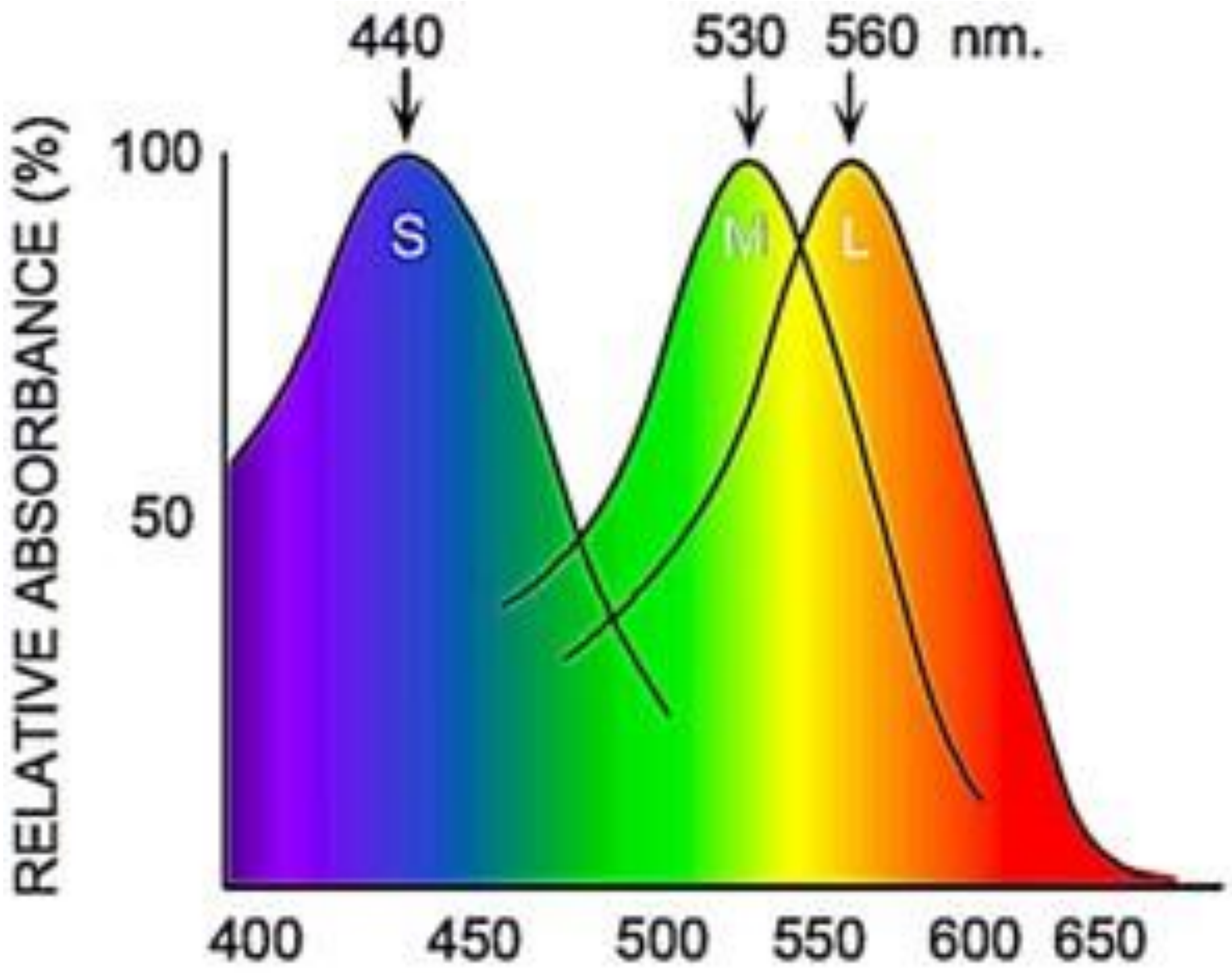
Rods and Cones





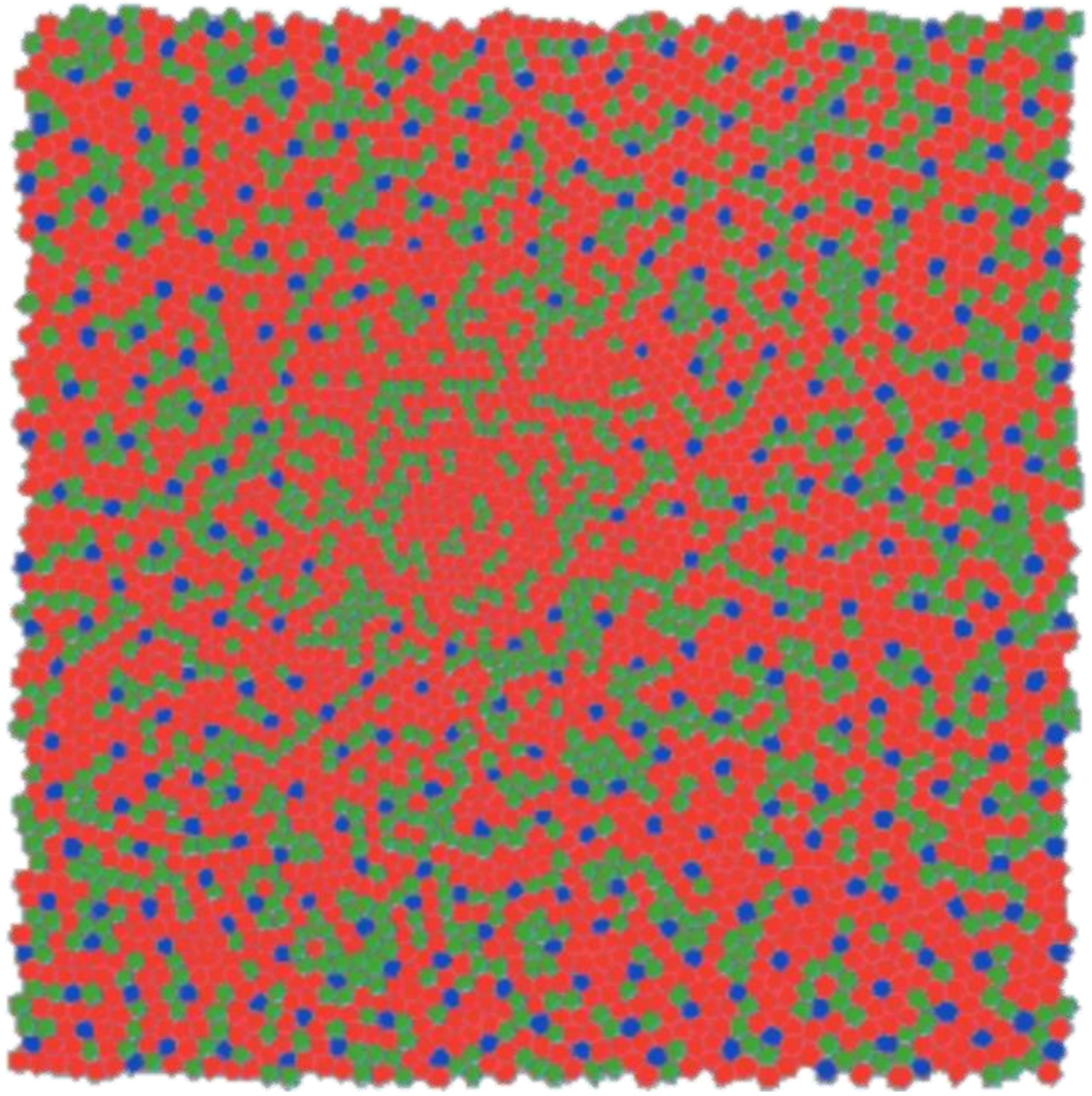
- $75 - 100 \cdot 10^6$ rods for perception of intensity
- $6 - 7 \cdot 10^6$ cones for perception of color
- Fovea
  - region in the center of the field of view
  - with high density of cones
  - but without rods

How Do We Perceive Colors? (1)

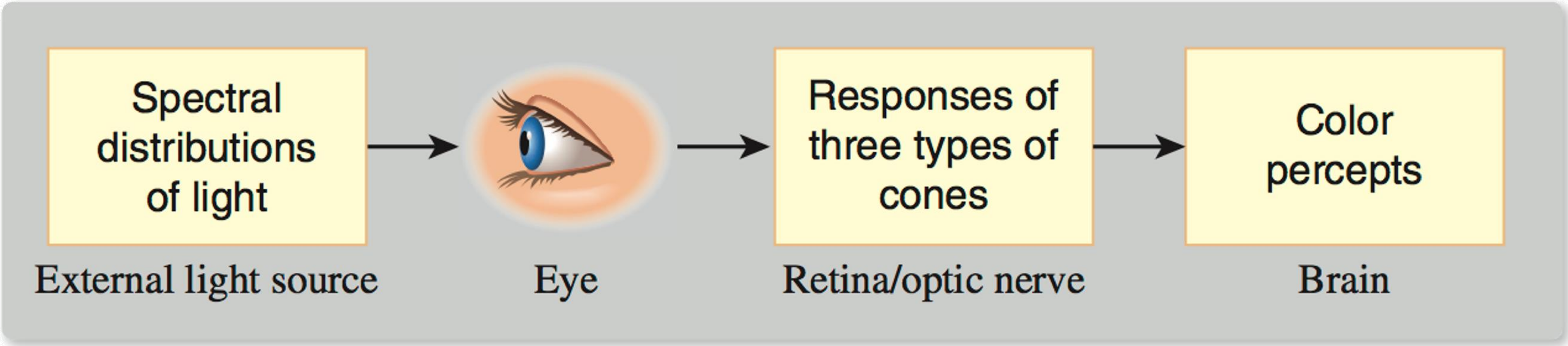


3 kinds of cones:

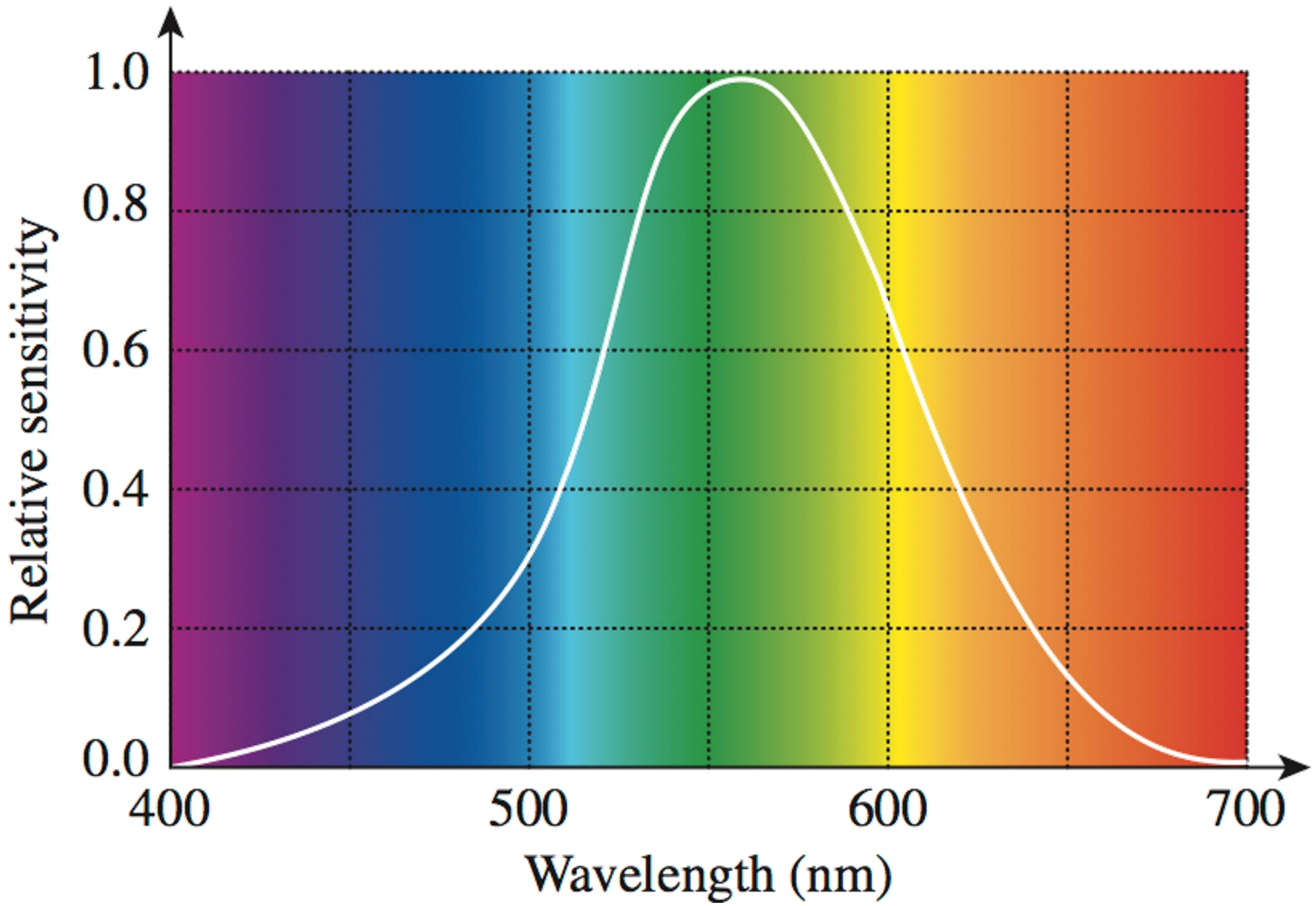
- S
  short wavelength
  455 nm
  blue
- M
  medium wavelength
  534 nm
  green
- L
  long wavelength
  563 nm
  red



How Do We Perceive Colors? (2)

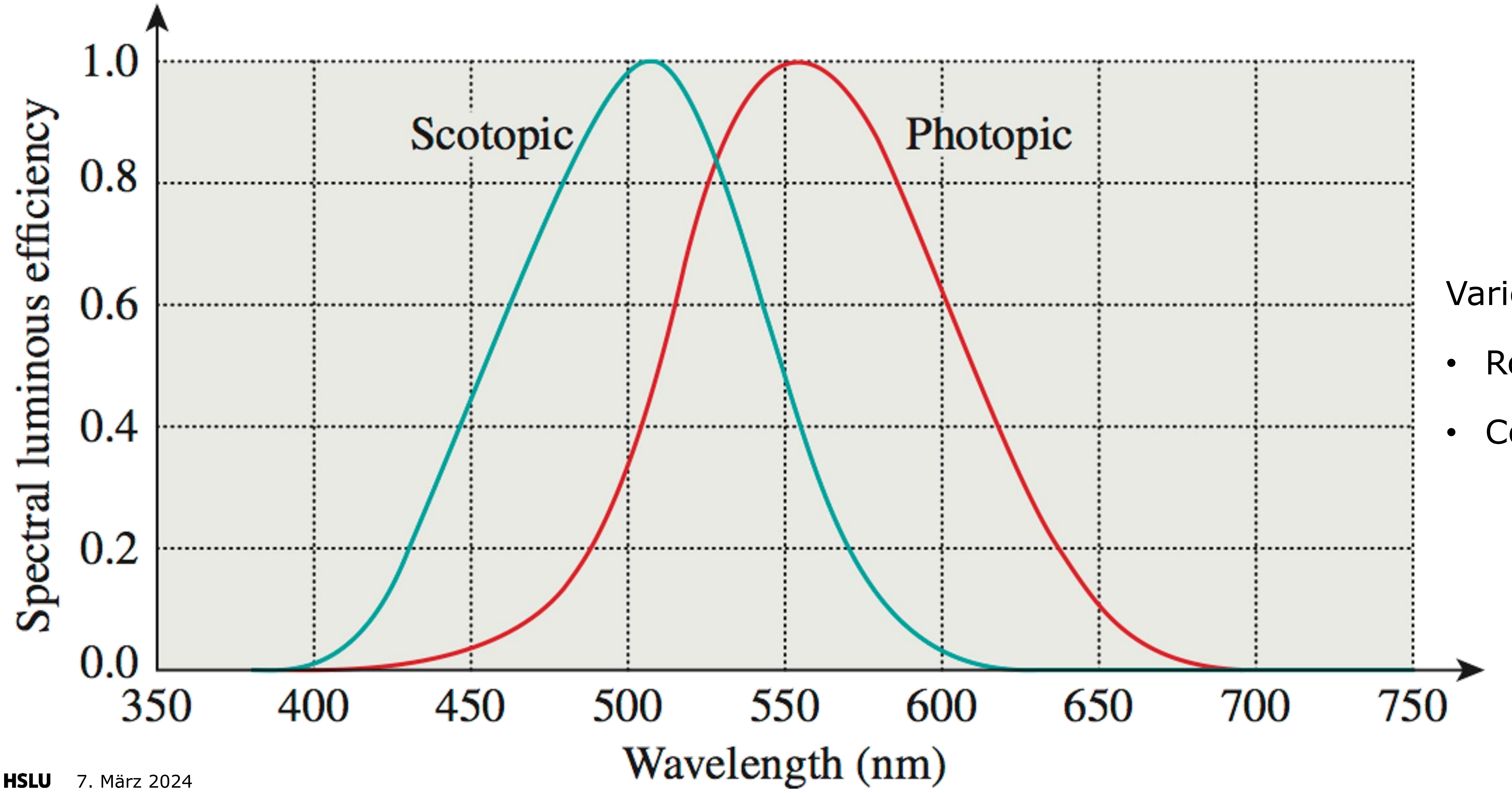


Perception of Brightness (1)



Experiment on perception of brightness

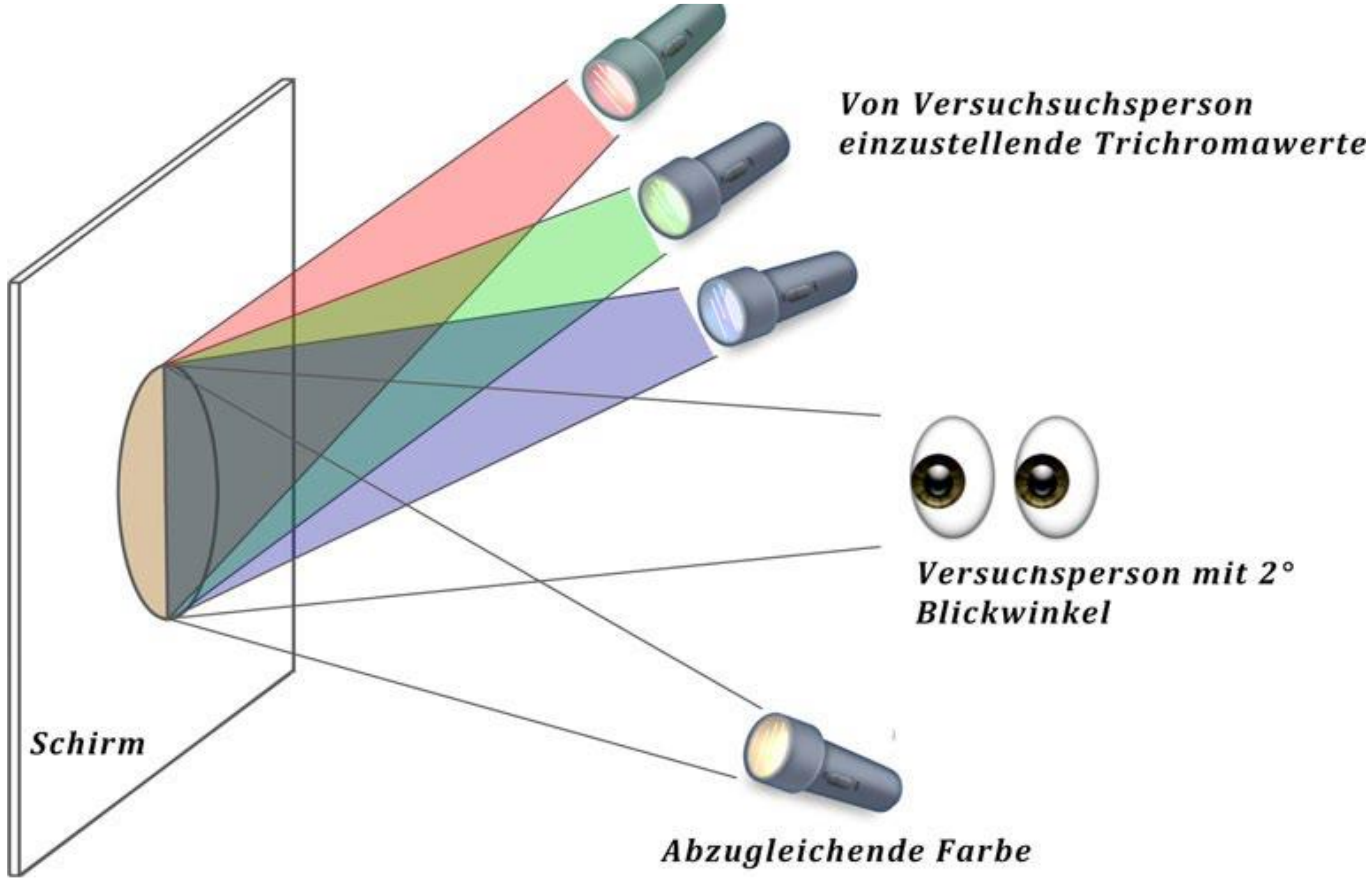
Perception of Brightness (2)



Various brightness sensitivity

- Rods: Scotopic vision
- Cones: Photopic vision

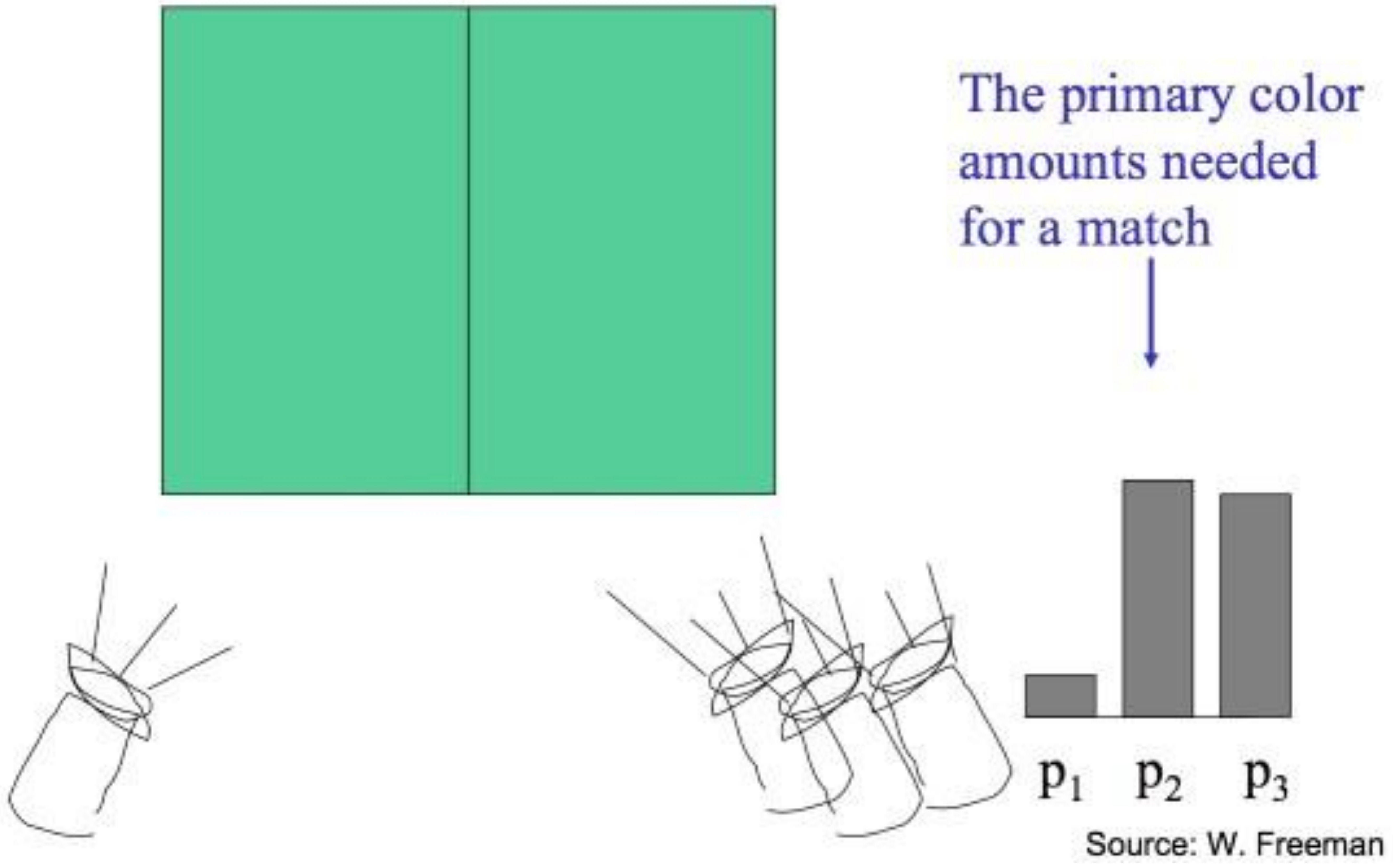
Experiment: Mixing Primary Colors (1)



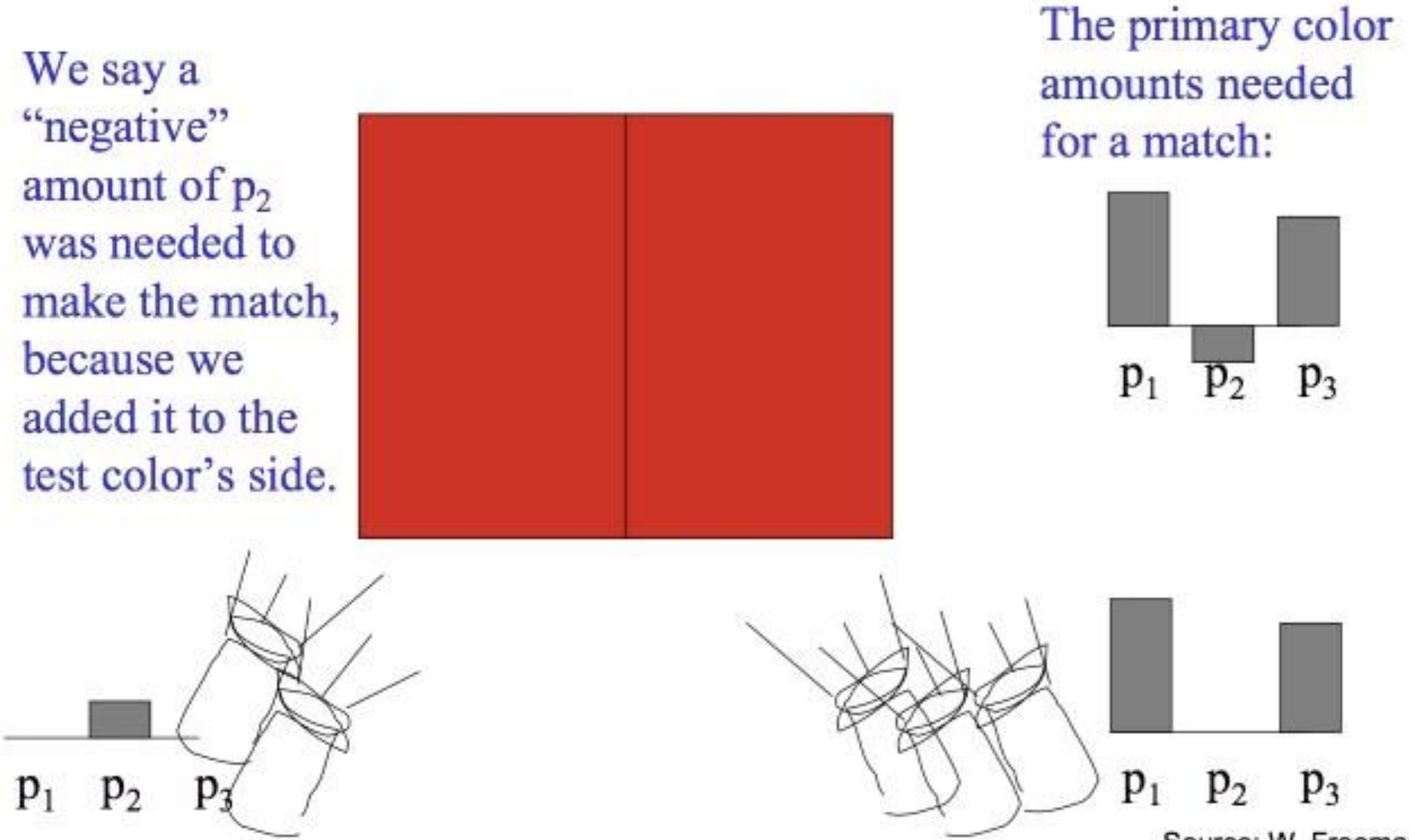
Is it possible

- to mix all colors
- using the 3 primary colors?

Experiment: Mixing Primary Colors (2)

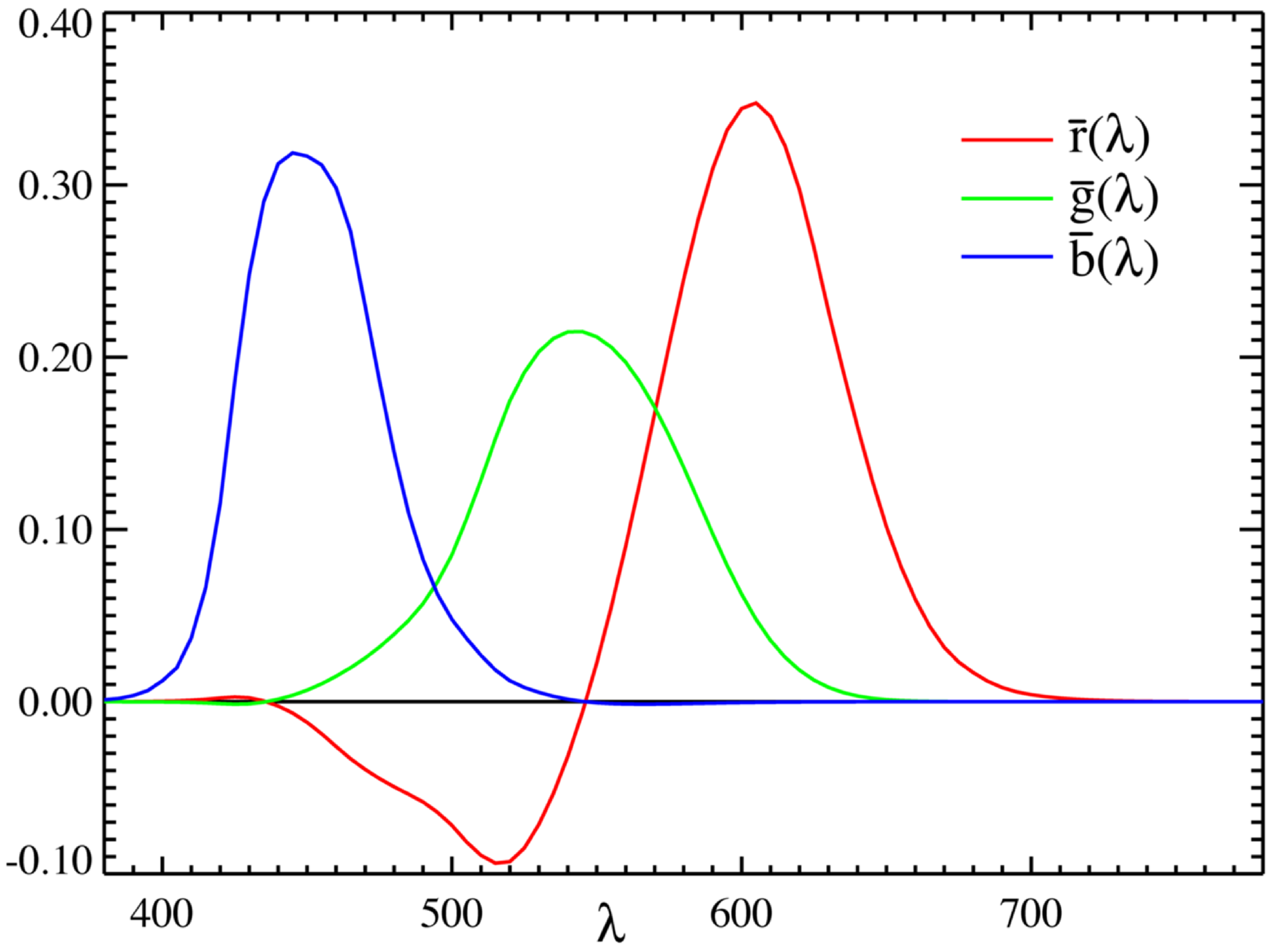


Experiment: Mixing Primary Colors (3)



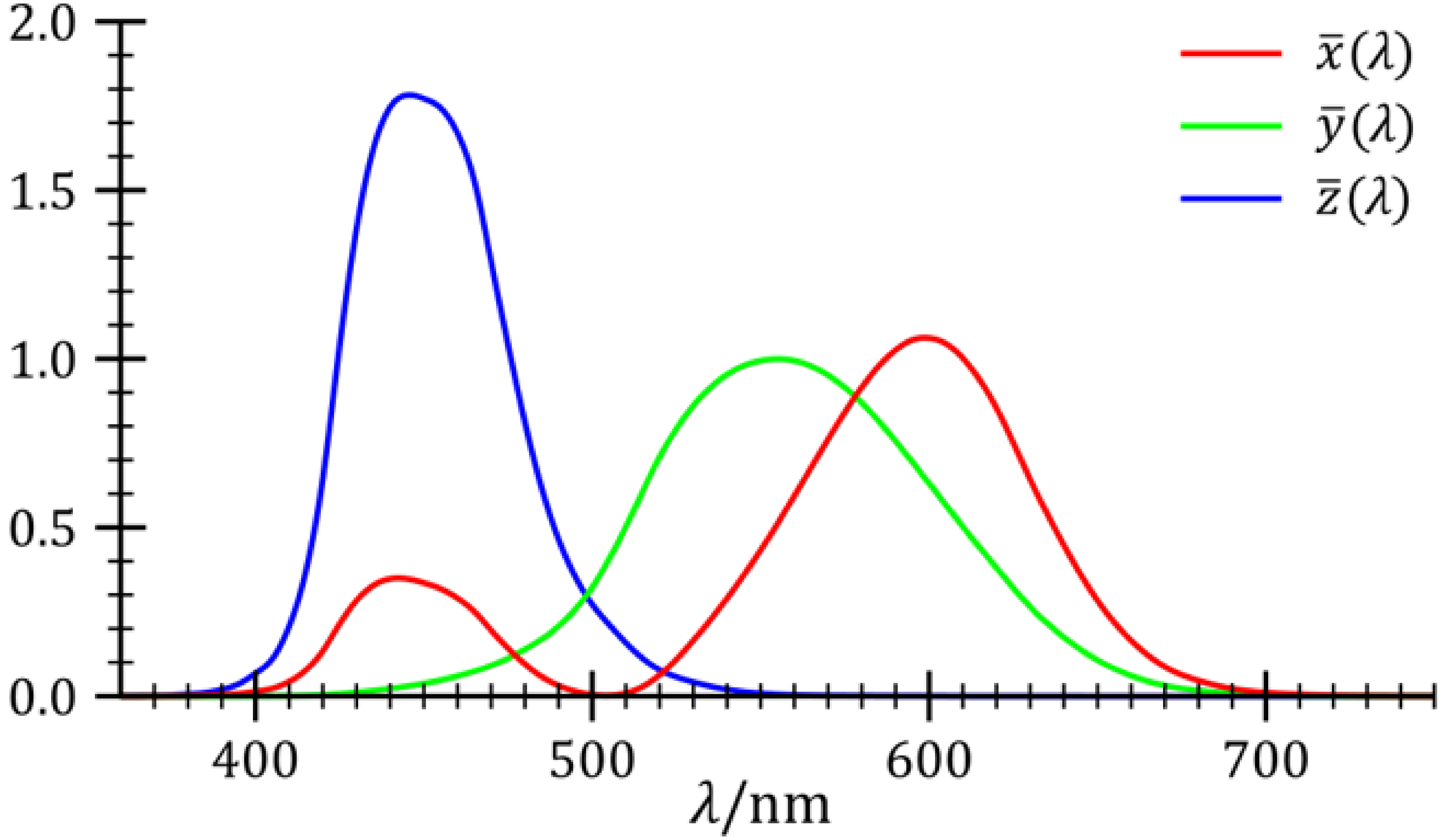
Source: W. Freeman

Experiment: Mixing Primary Colors (4)



Result: The CIE RGB color matching functions
(CIE: Commission internationale de l'éclairage)

CIE Color Space (1)



The CIE XYZ standard observer color matching functions
(transform RGB to XYZ to avoid negative values)

CIE Color Space (2)

Given an arbitrary color by its spectral distribution $P(\lambda)$,
its X, Y, and Z portions are given by

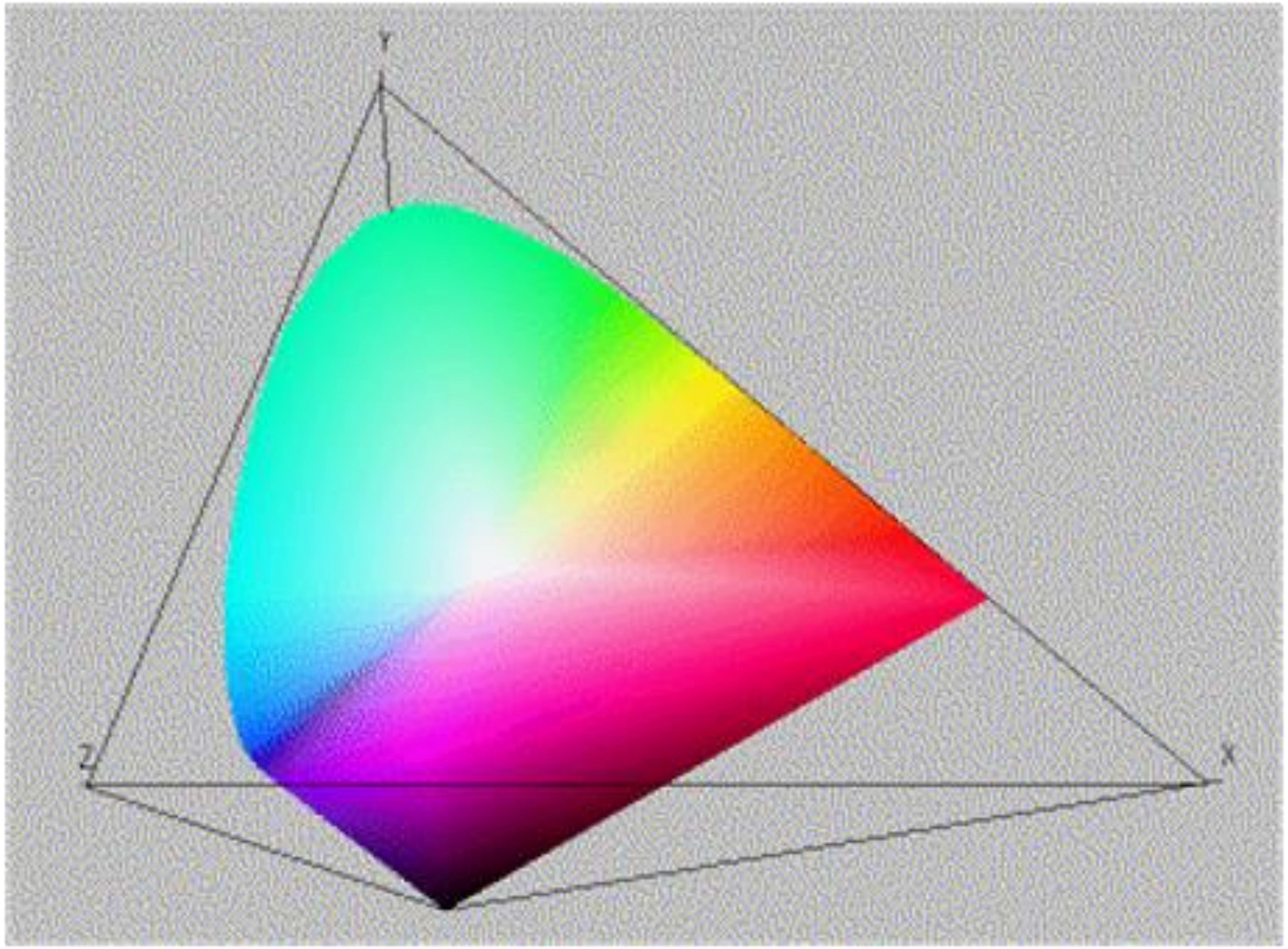
$$X = k \int_{\Lambda} P(\lambda)\,\bar{x}(\lambda)\,\mathrm{d}\lambda$$

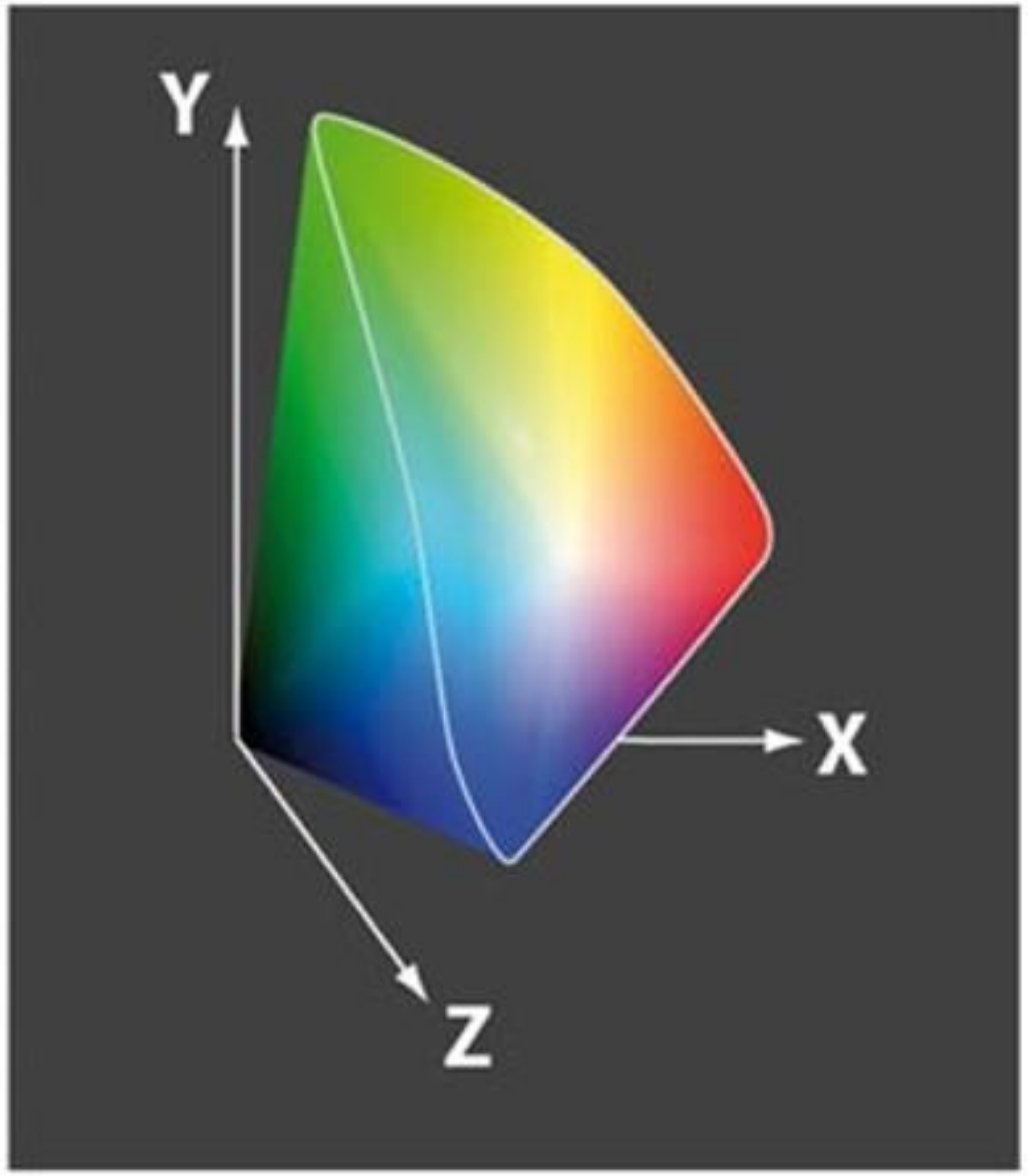
$$Y = k \int_{\Lambda} P(\lambda)\,\bar{y}(\lambda)\,\mathrm{d}\lambda$$

$$Z = k \int_{\Lambda} P(\lambda)\,\bar{z}(\lambda)\,\mathrm{d}\lambda$$

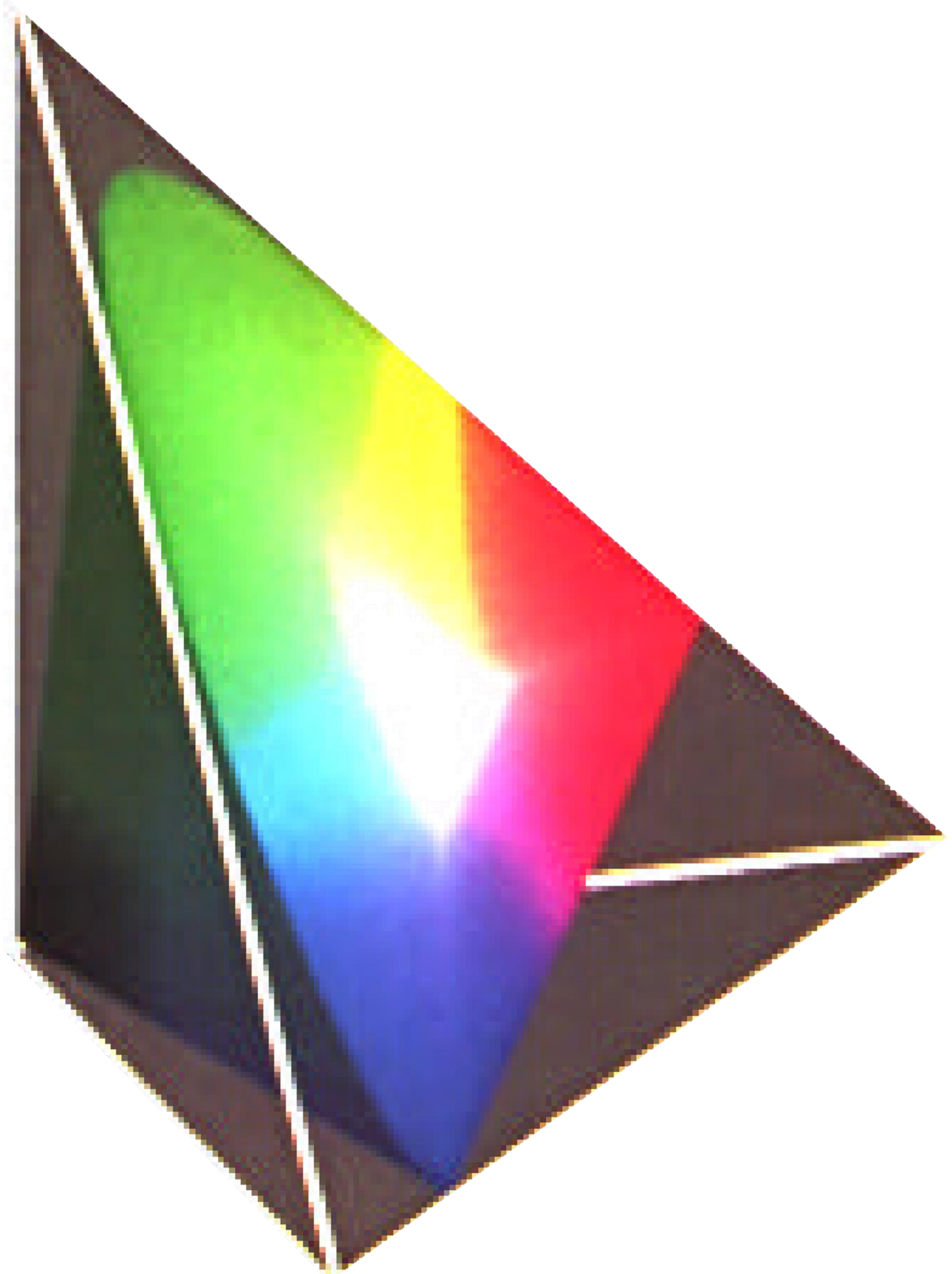
$$\Lambda = [380\ \mathrm{nm}, 780\ \mathrm{nm}]$$

CIE Color Space (3)



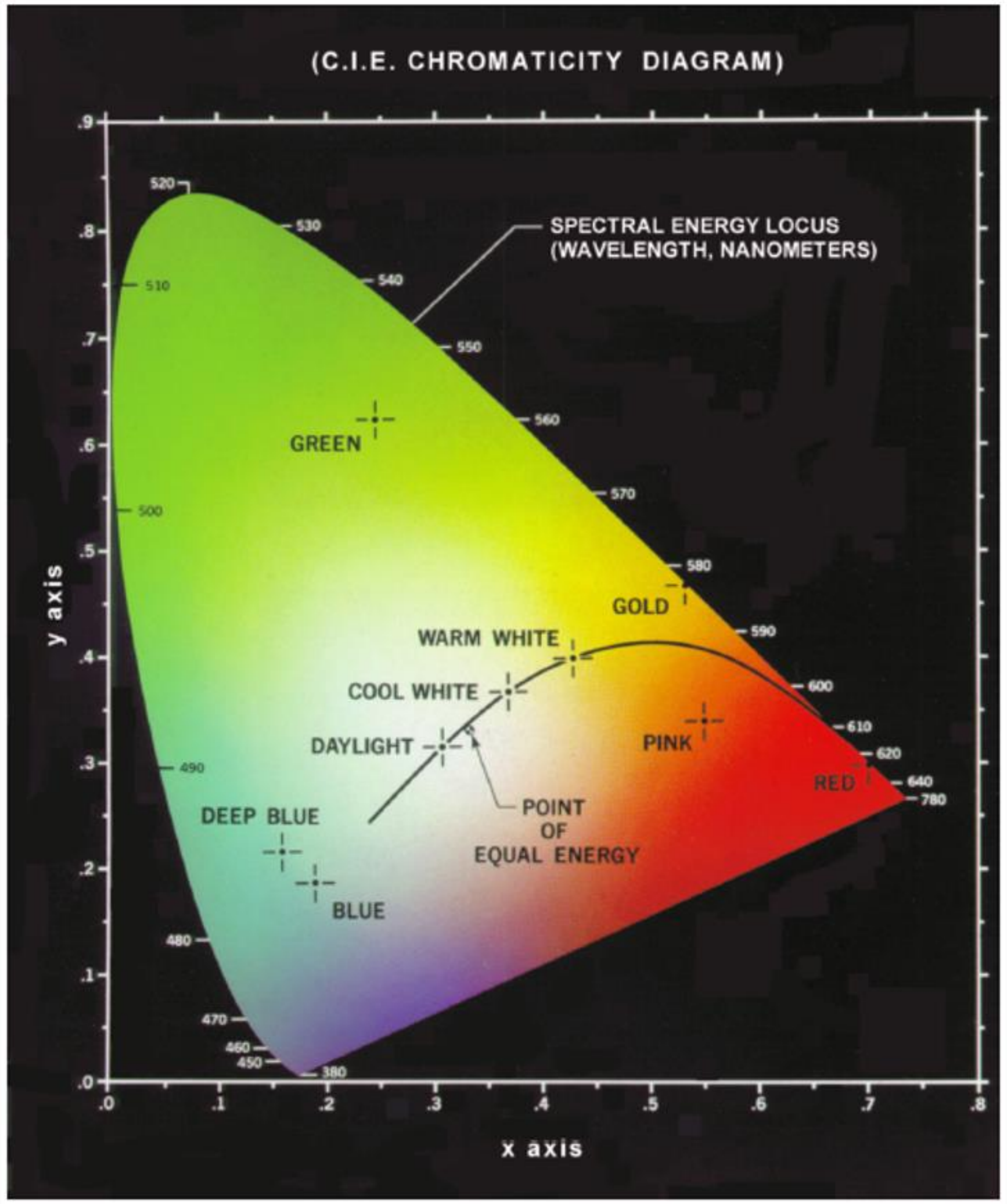


CIE Color Space (4)



- chromaticity diagram
  colors on plane through points (1, 0, 0), (0, 1, 0), and (0, 0, 1) in XYZ-space

CIE Color Space (4)

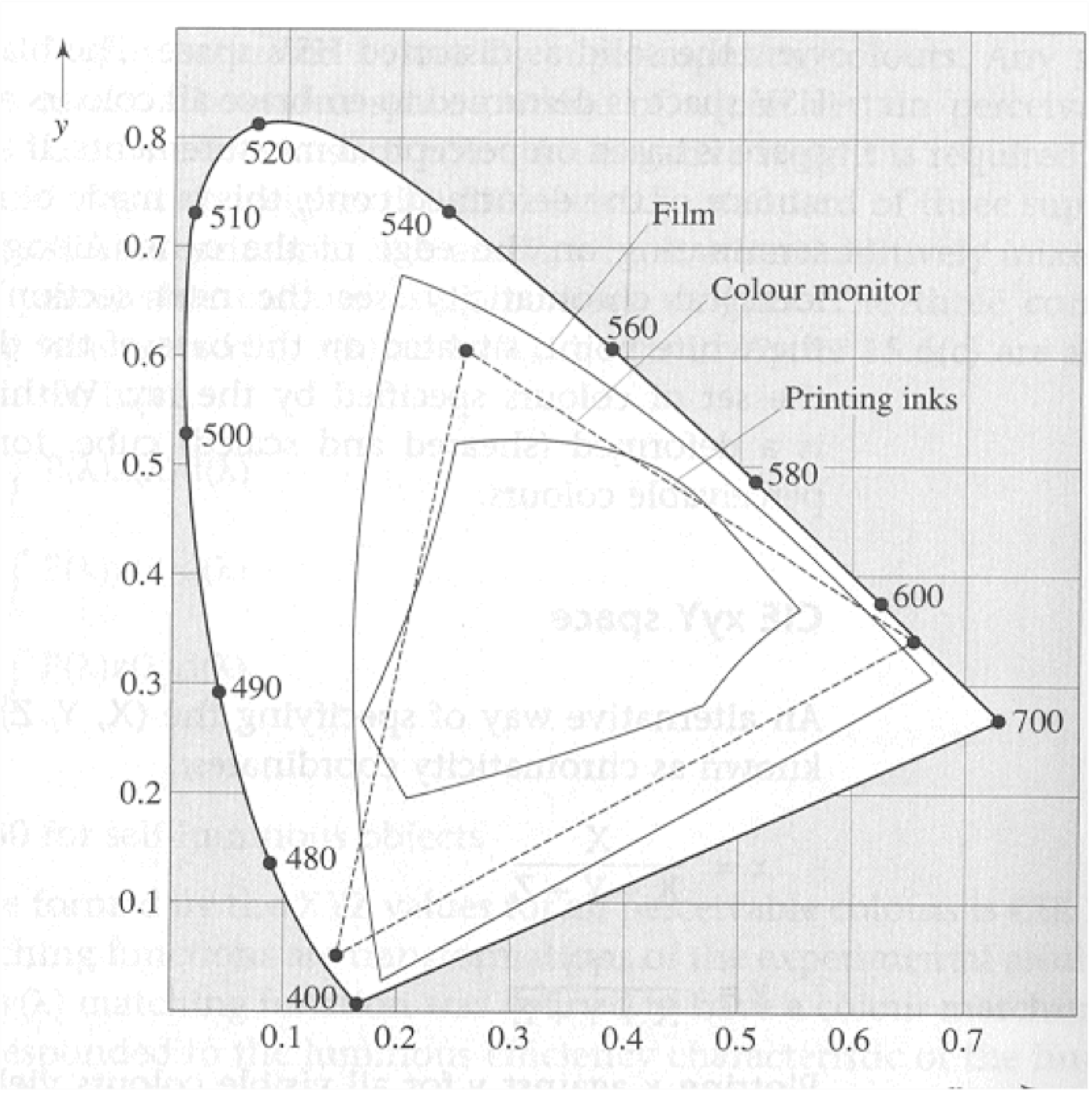


- spectral colors
  on boundary of chromaticity diagram
- mixing two colors
  using linear interpolation
- purple boundary
  non-spectral colors, mixed from colors violet (380 nm) and red (780 nm)
- white point
- dominant wavelength
  ray from white point through given color intersects boundary
- complementary wavelength
  ray from dominant wavelength through white point intersects boundary
- complementary color
  reflection at white point

Exercise 1

Quiz 1 on Ilias

Which Colors Can Various Devices Display?



Color Models

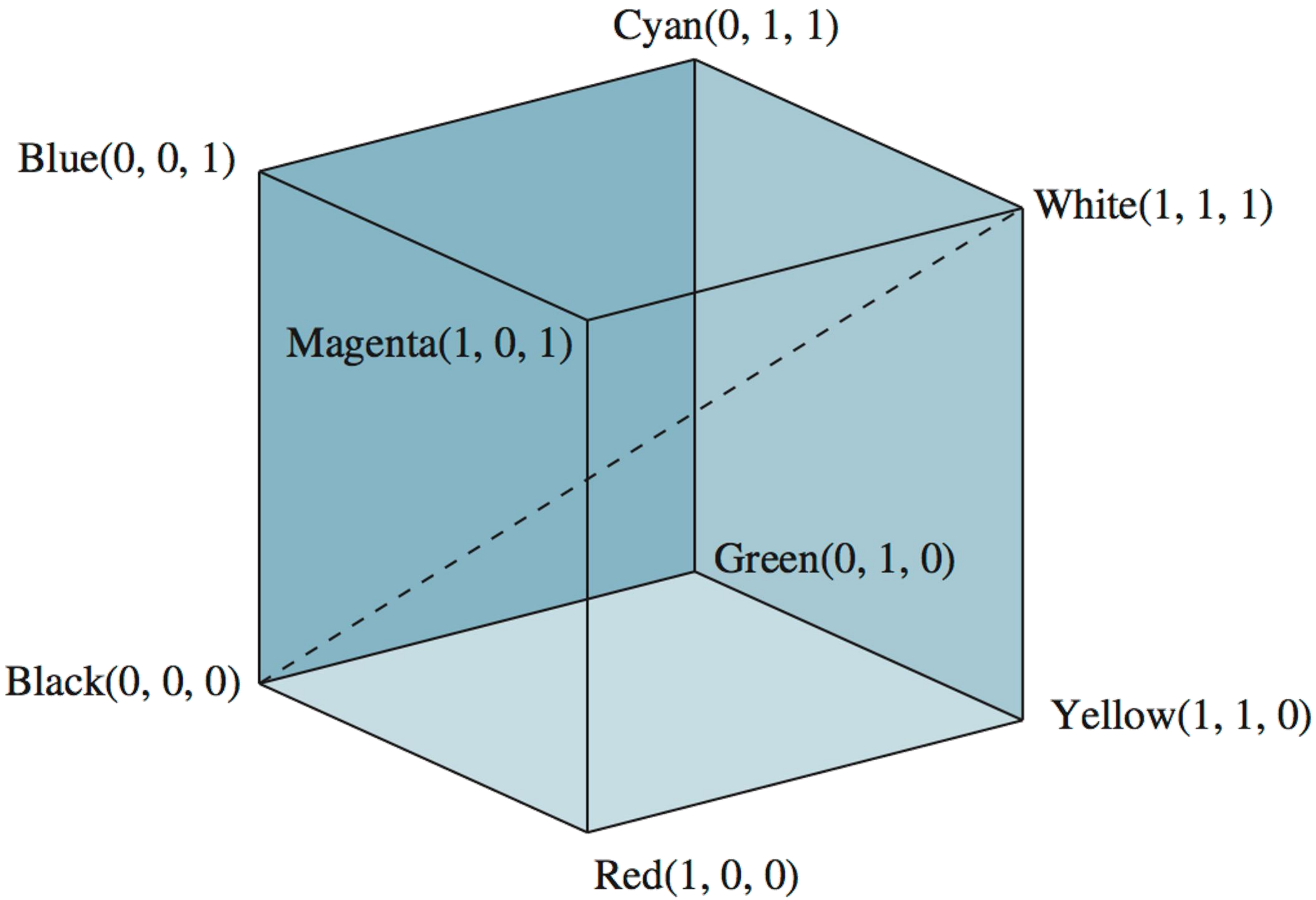
Hardware oriented

- RGB for computer monitors
  R: red
  G: green
  B: blue
- CMY and CMYK for color printing
  C: cyan
  M: magenta
  Y: yellow
  K: key (black)
- YIQ for television
  Y: luminance
  I: difference between cyan and orange
  Q: difference between magenta and green

User oriented

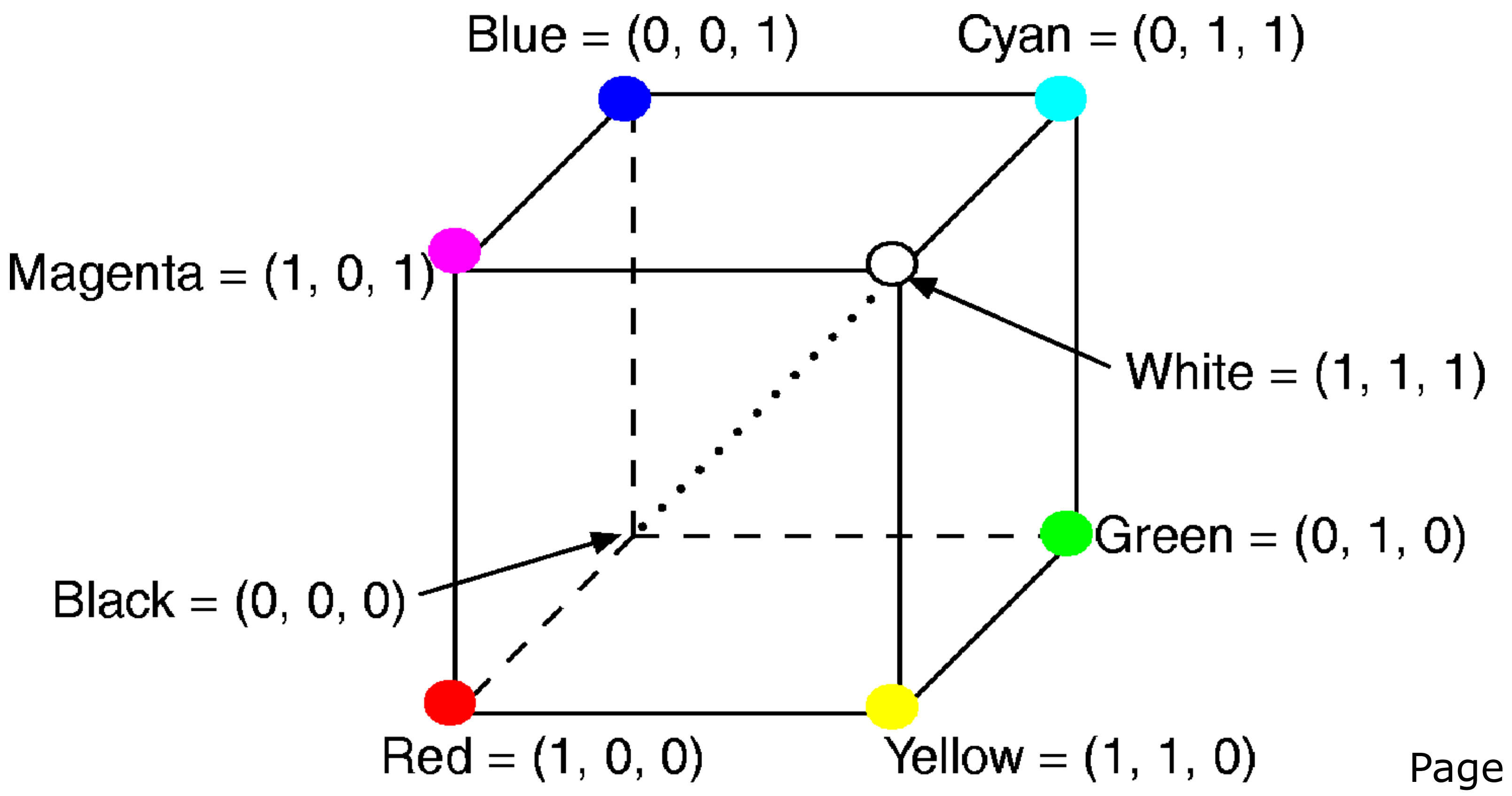
- HSV
  H: Hue
  S: Saturation
  V: Value
- Munsell system
- CIE Lab

RGB Color Model (1)

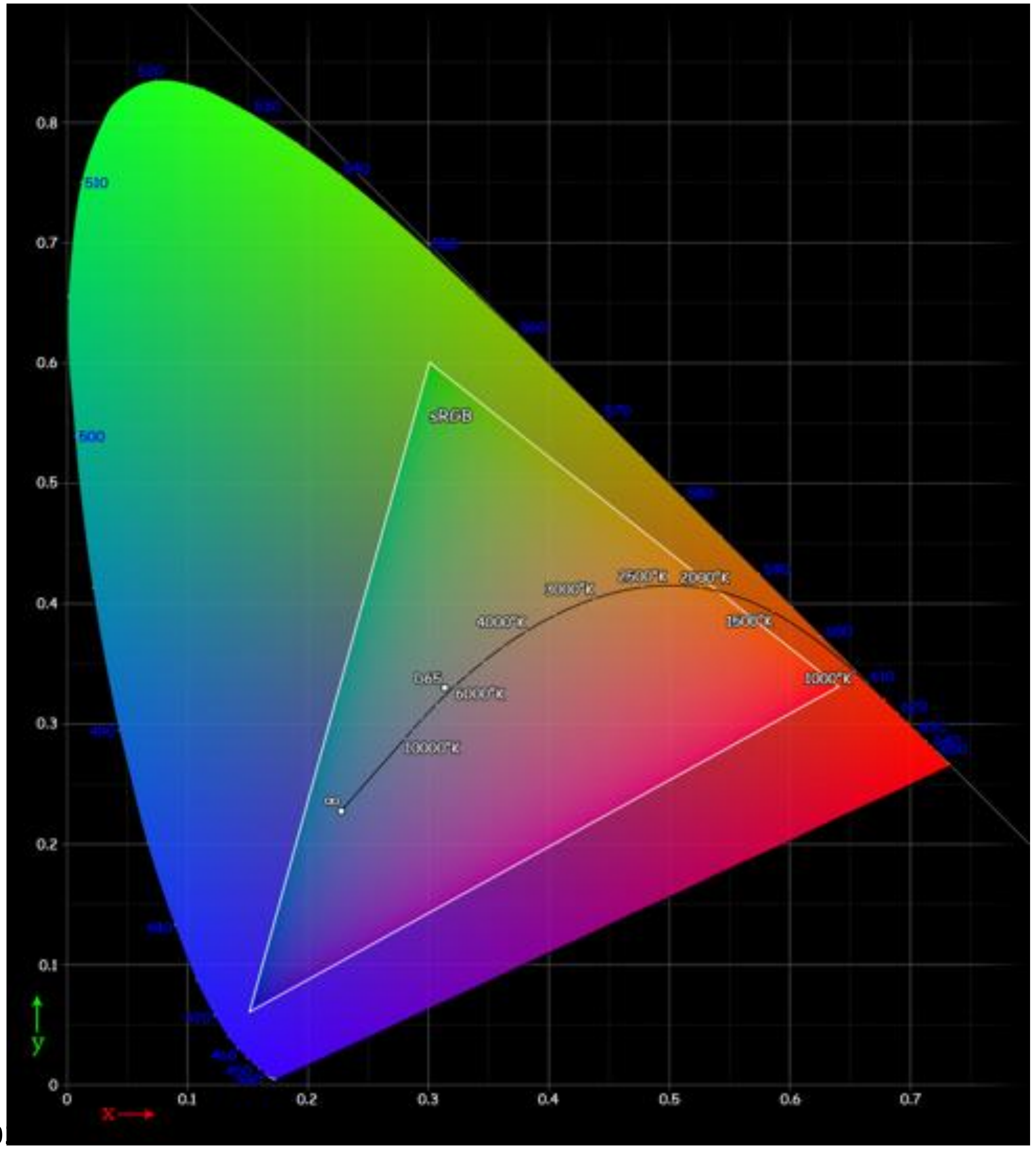


RGB Color Model (2)

- additive color model
- primary colors
  - R: red
  - G: green
  - B: blue
- colors: (r, g, b)
  $(r, g, b) \in [0,1] \times [0,1] \times [0,1] = [0,1]^3$

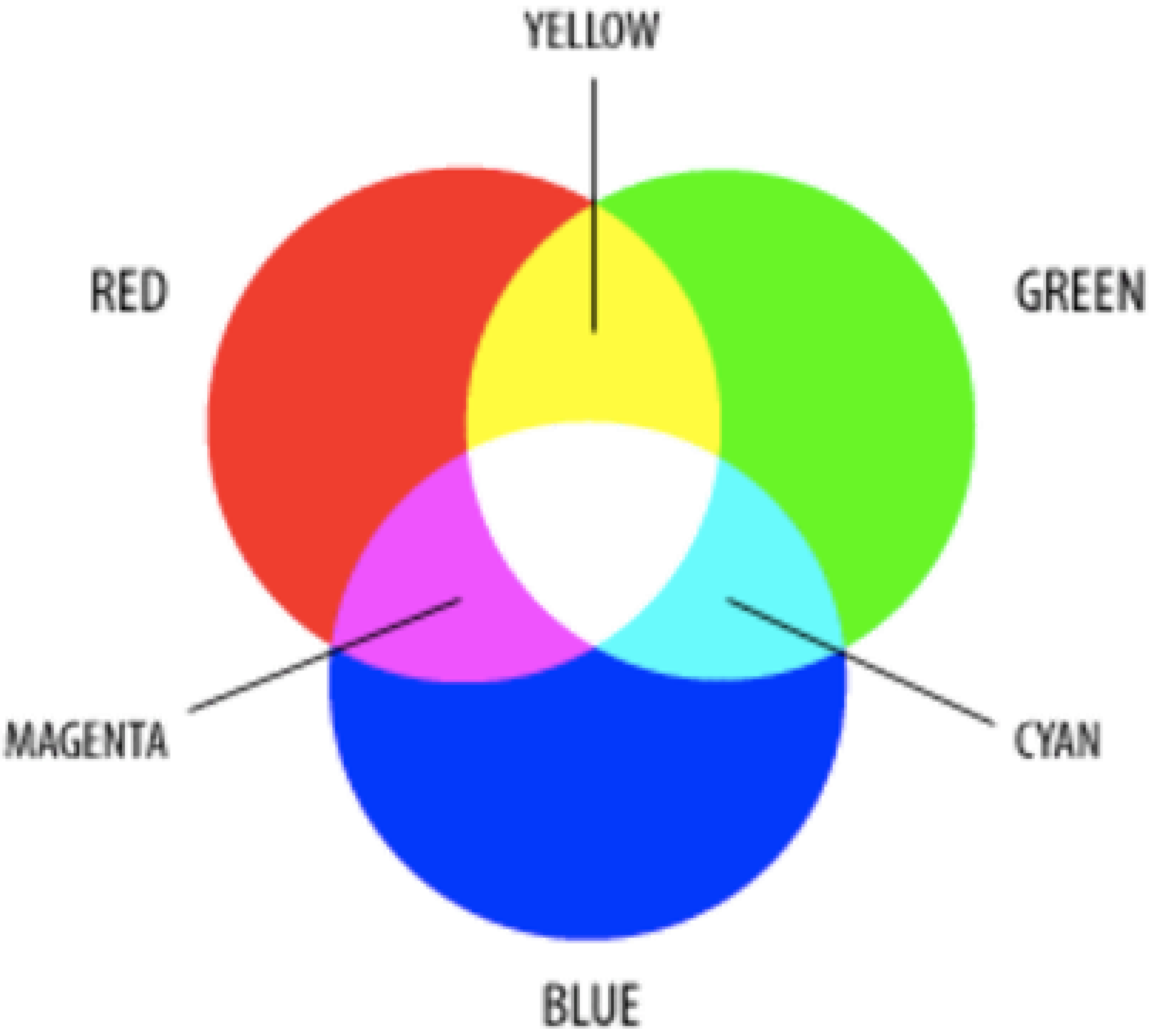


sRGB Color Space in CIE XYZ



- s: standard
- created by
  - Hewlett-Packard
  - Microsoft
- used on
  - monitors
  - printers
  - world wide web
- standardized as as IEC 61966-2-1:1999

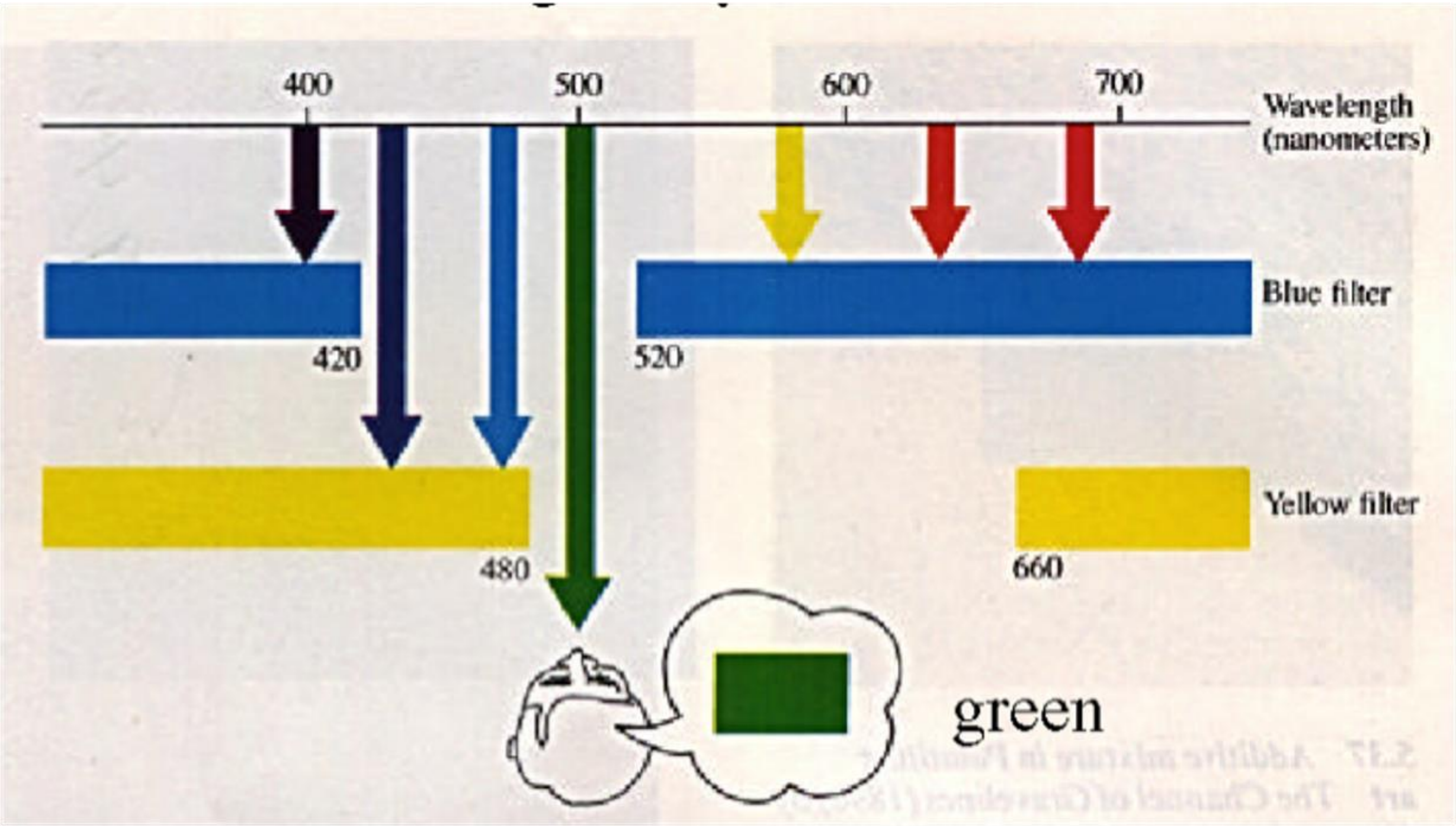
Additive Color Mixing (1)



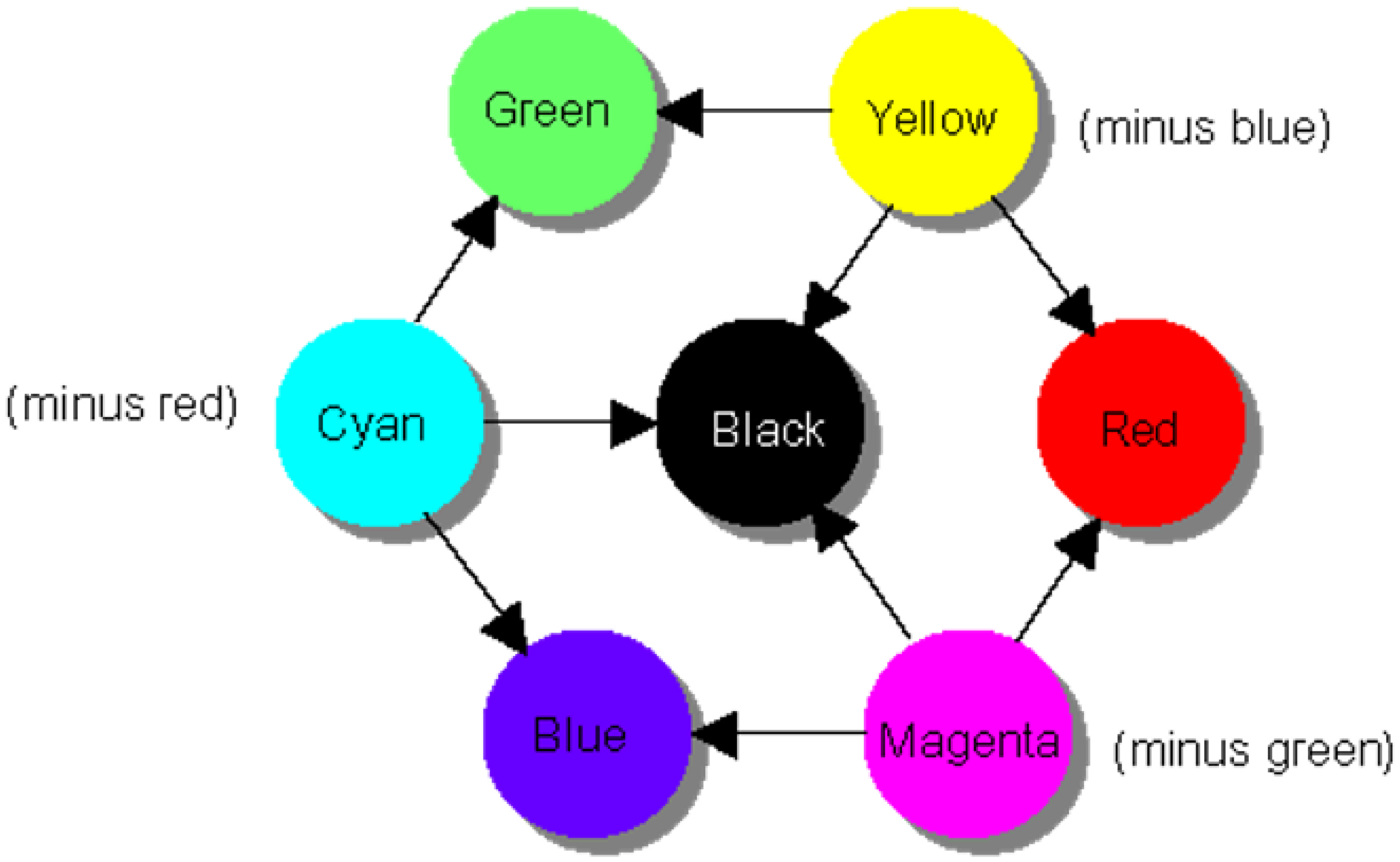
Additive Color Mixing (2)

- Pointillism
- Artists
  - Georges Seurat
  - Gustave Cariot
  - Paul Signac
  - Henri Edmond Cross
  - Giovanni Segantini
  - Camille Pissarro

Subtractive Color Mixing



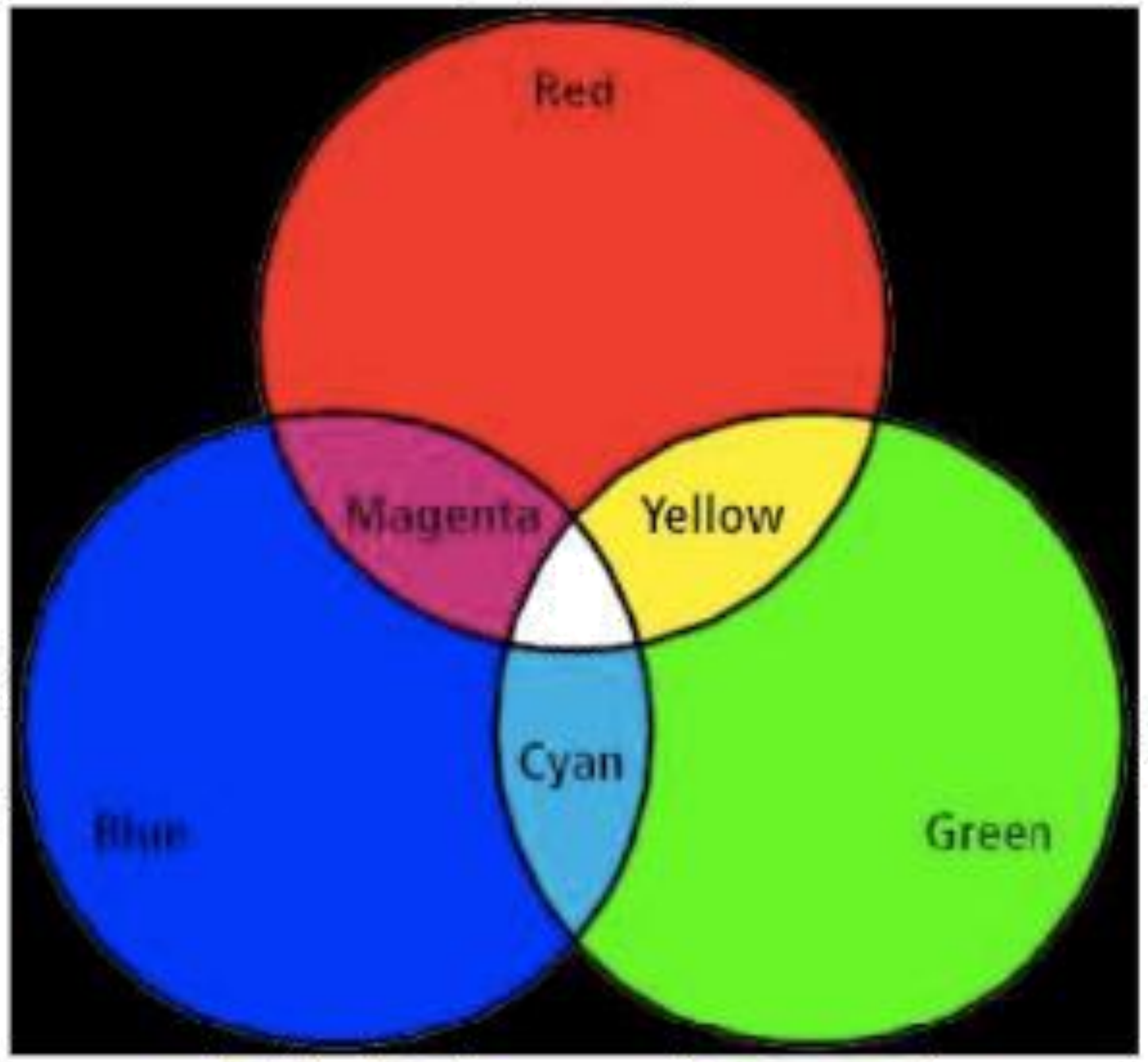
CMY Color Model



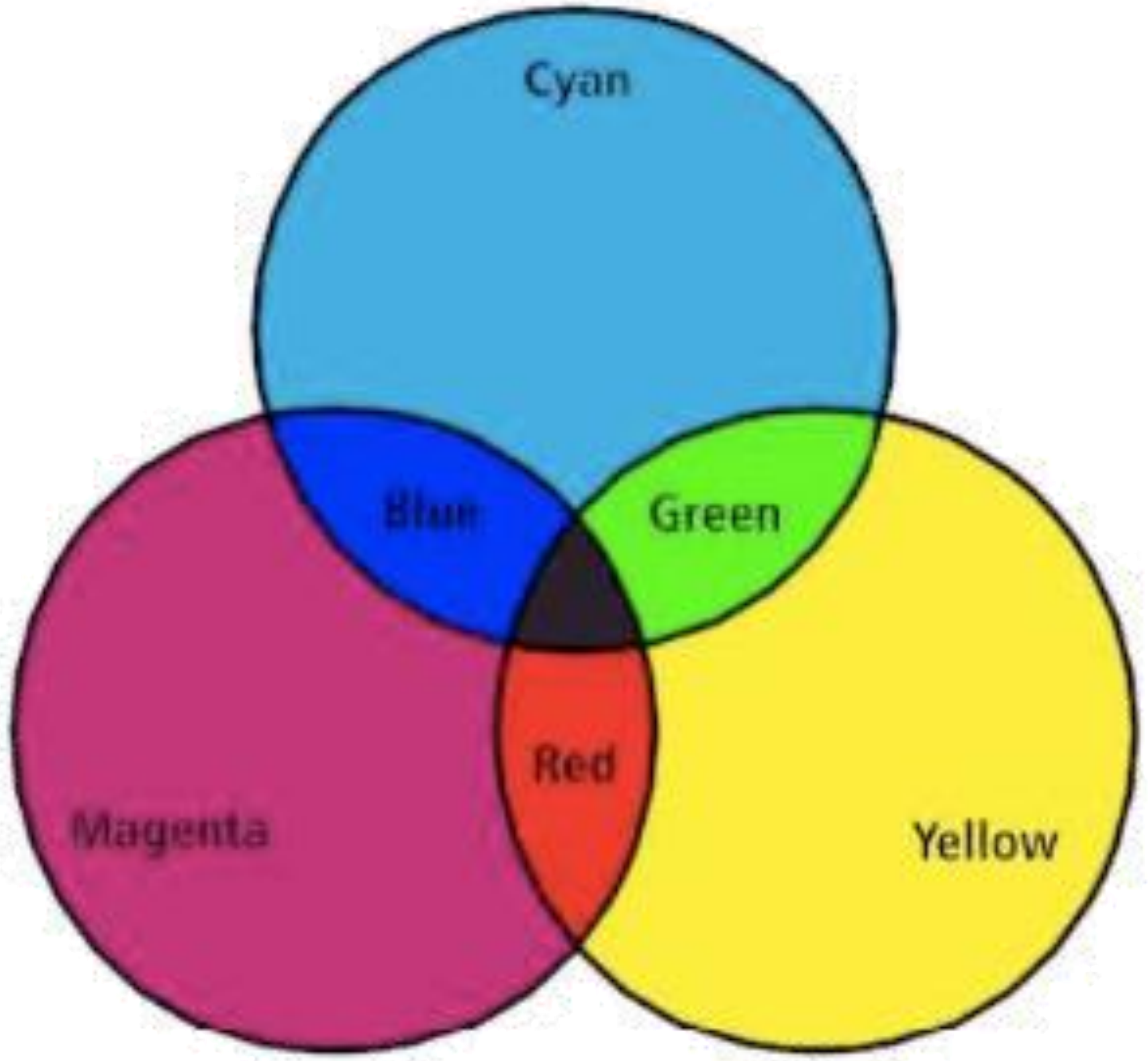
- subtractive color model
- primary colors
  - C: cyan
  - M: magenta
  - Y: yellow
- interrelation to RGB

$$\begin{pmatrix} C \\ M \\ Y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} R \\ G \\ B \end{pmatrix}$$

Additive vs Subtractive Color Mixing

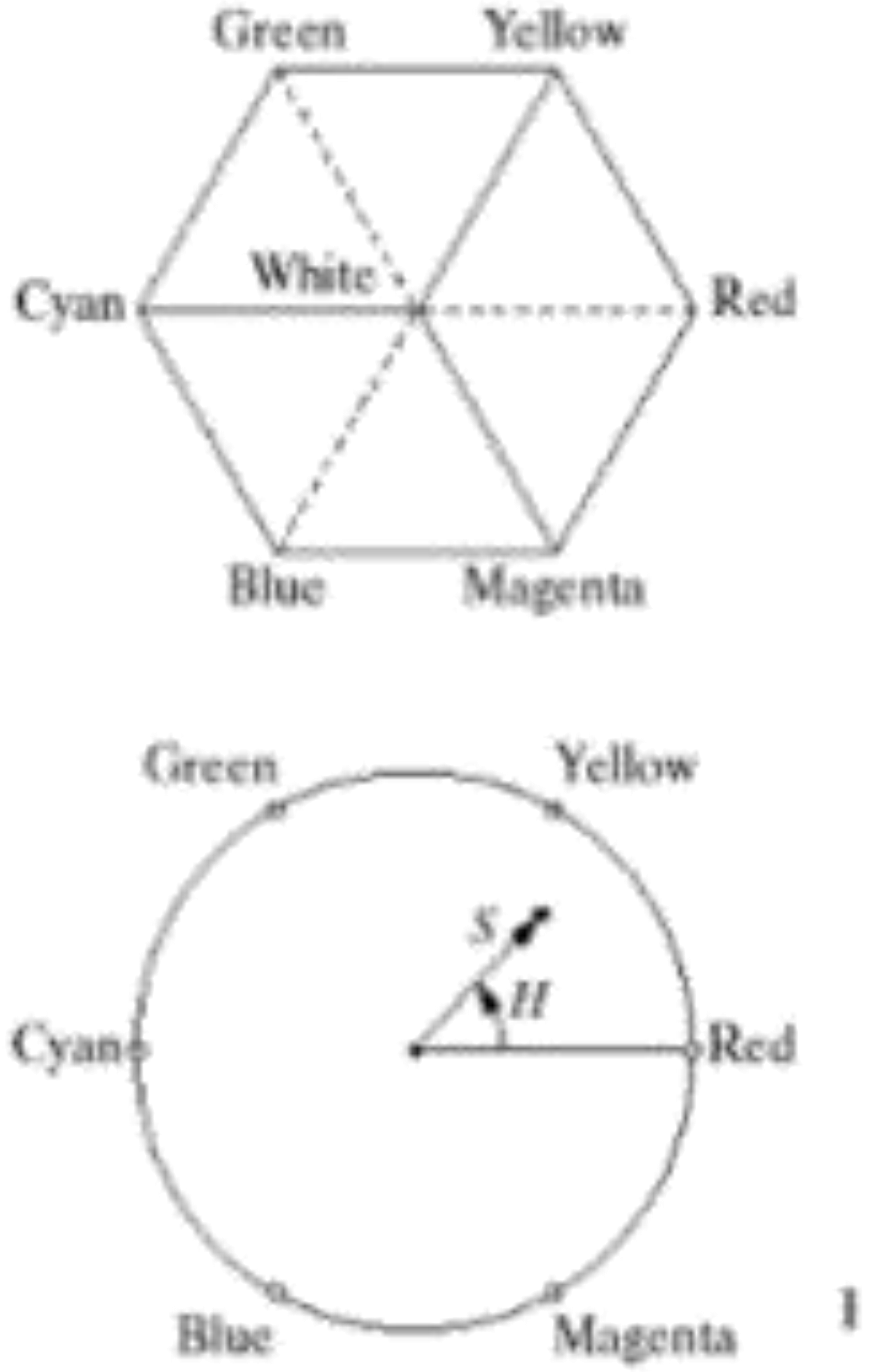


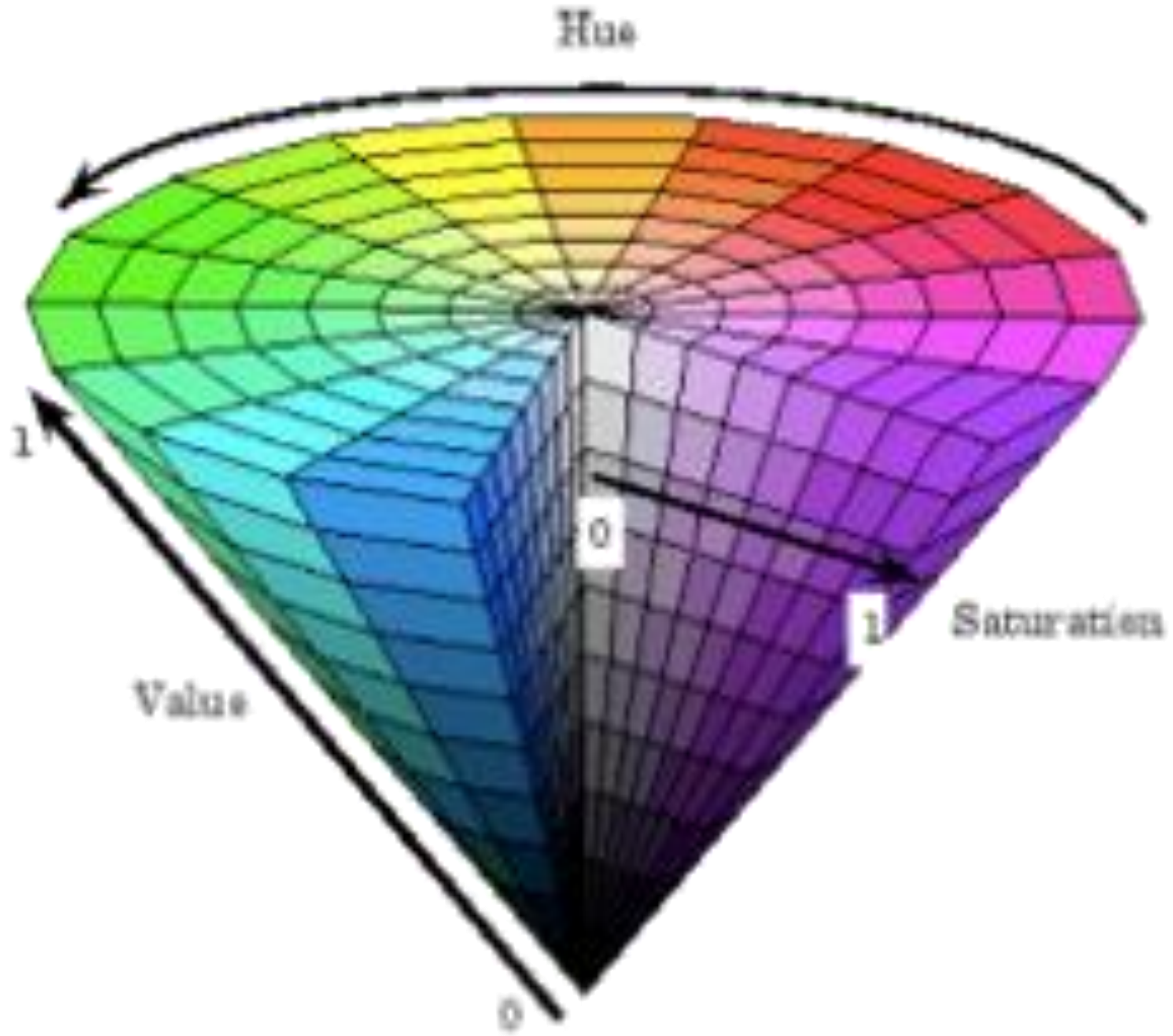
Magenta = Red + Blue
Cyan = Blue + Green
Yellow = Green + Red

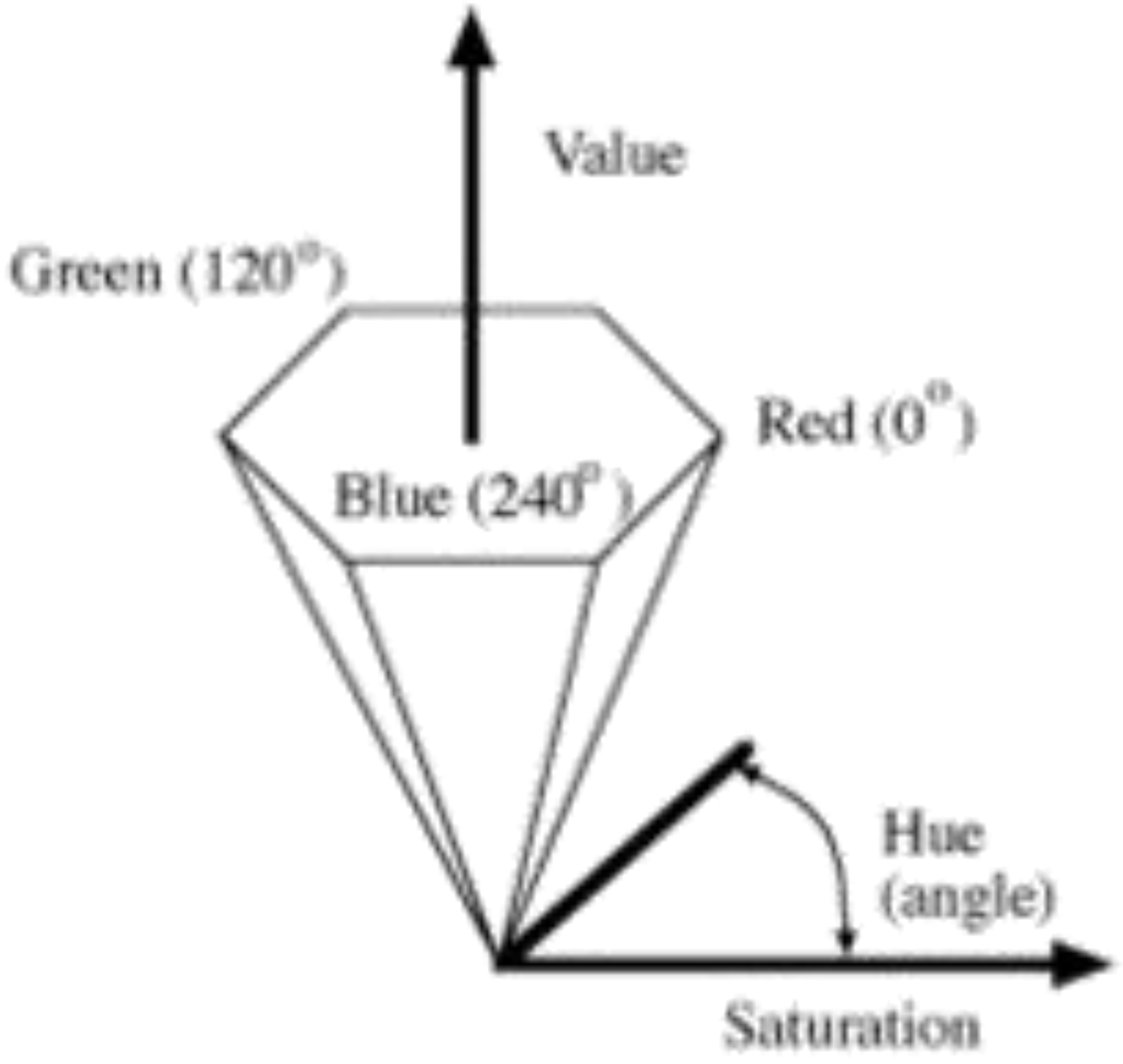


Magenta = White - Green
Cyan = White - Red
Yellow = White - Blue

HSV Color Model





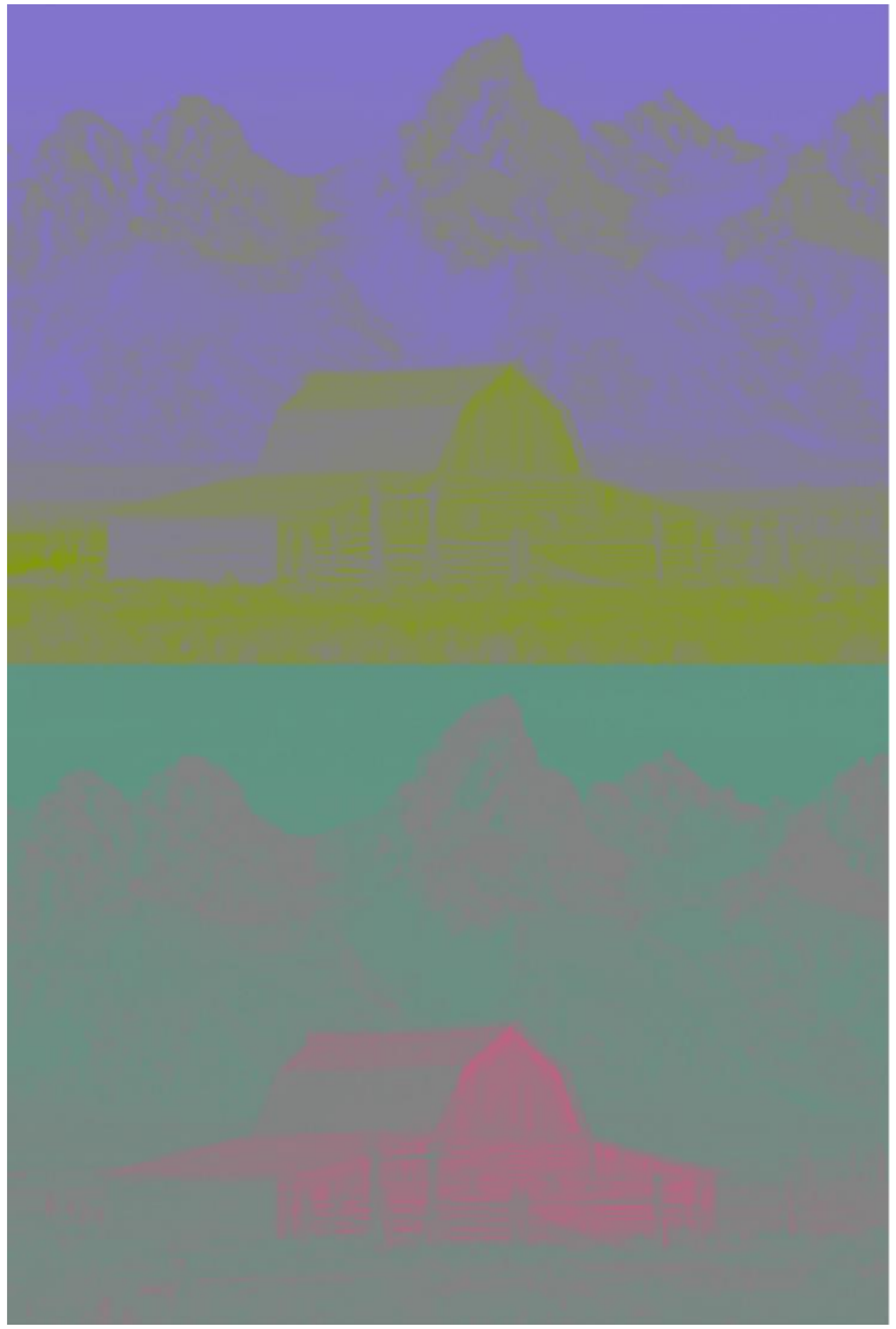


- intuitive
- color attributes
  - H: hue
  - S: saturation
  - V: value (intensity)

Examples of HSV Colors

Farbe	H	S	V	R	G	B
Schwarz	–	–	0 %	0 %	0 %	0 %
Rot	0°	100 %	100 %	100 %	0 %	0 %
Gelb	60°	100 %	100 %	100 %	100 %	0 %
Braun	24,3°	75 %	36,1 %	36 %	20 %	9 %
Weiß	–	0 %	100 %	100 %	100 %	100 %
Grün	120°	100 %	100 %	0 %	100 %	0 %
Dunkelgrün	120°	100 %	50 %	0 %	50 %	0 %
Cyan	180°	100 %	100 %	0 %	100 %	100 %
Blau	240°	100 %	100 %	0 %	0 %	100 %
Magenta	300°	100 %	100 %	100 %	0 %	100 %
Orange	30°	100 %	100 %	100 %	50 %	0 %
Violett	270°	100 %	100 %	50 %	0 %	100 %

Composite Video (YUV)

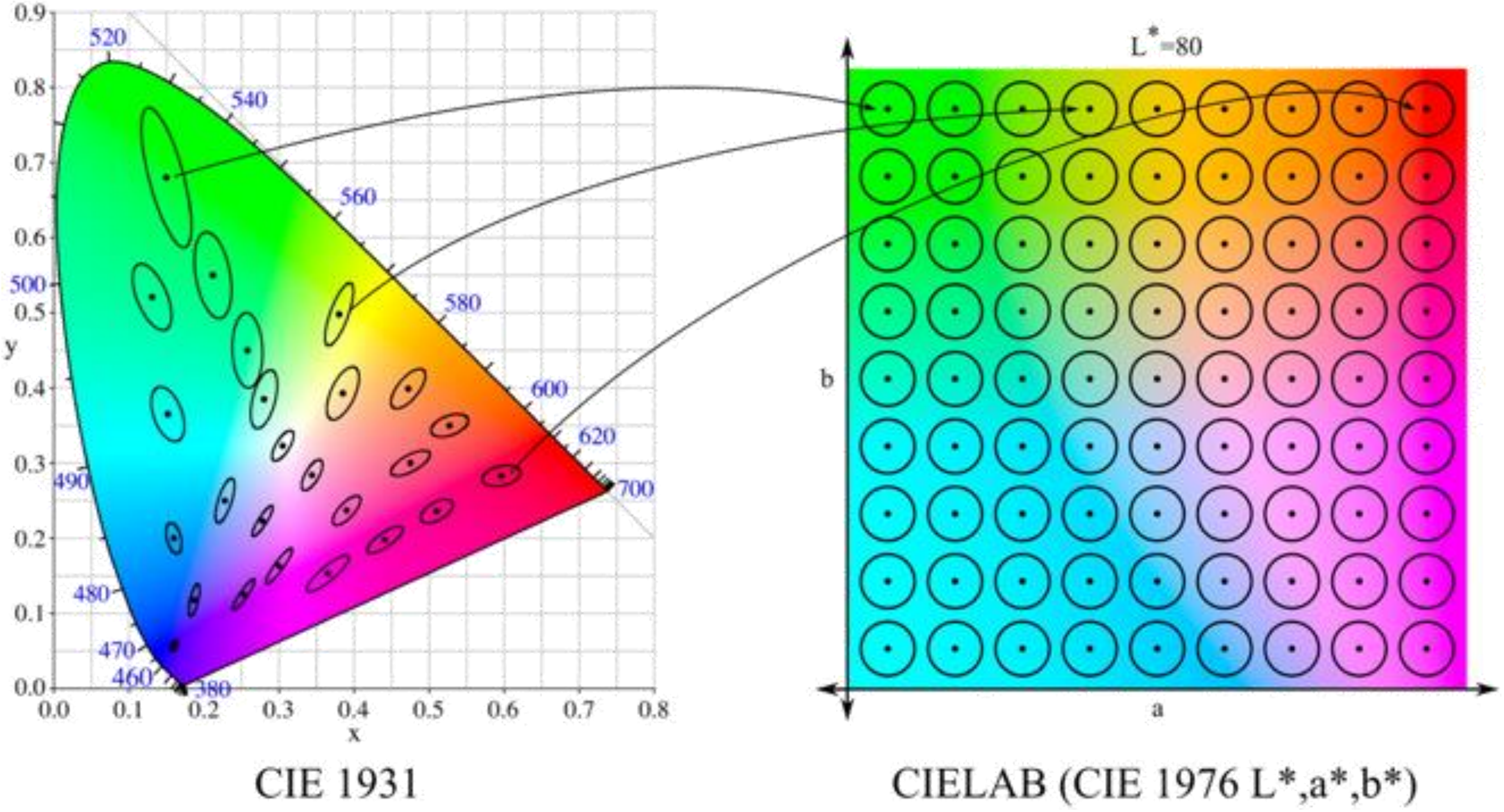


- Attributes:
  - Y: luminance
  - UV: chrominance
    - combination of hue and saturation
    - in many cases a quarter of the resolution
  - interrelation to RGB

$$Y = 0.299 \cdot R + 0.587 \cdot G + 0.114 \cdot B$$
$$U = \frac{0.436 \cdot (B - Y)}{1 - 0.114}$$
$$V = \frac{0.615 \cdot (R - Y)}{1 - 0.299}$$

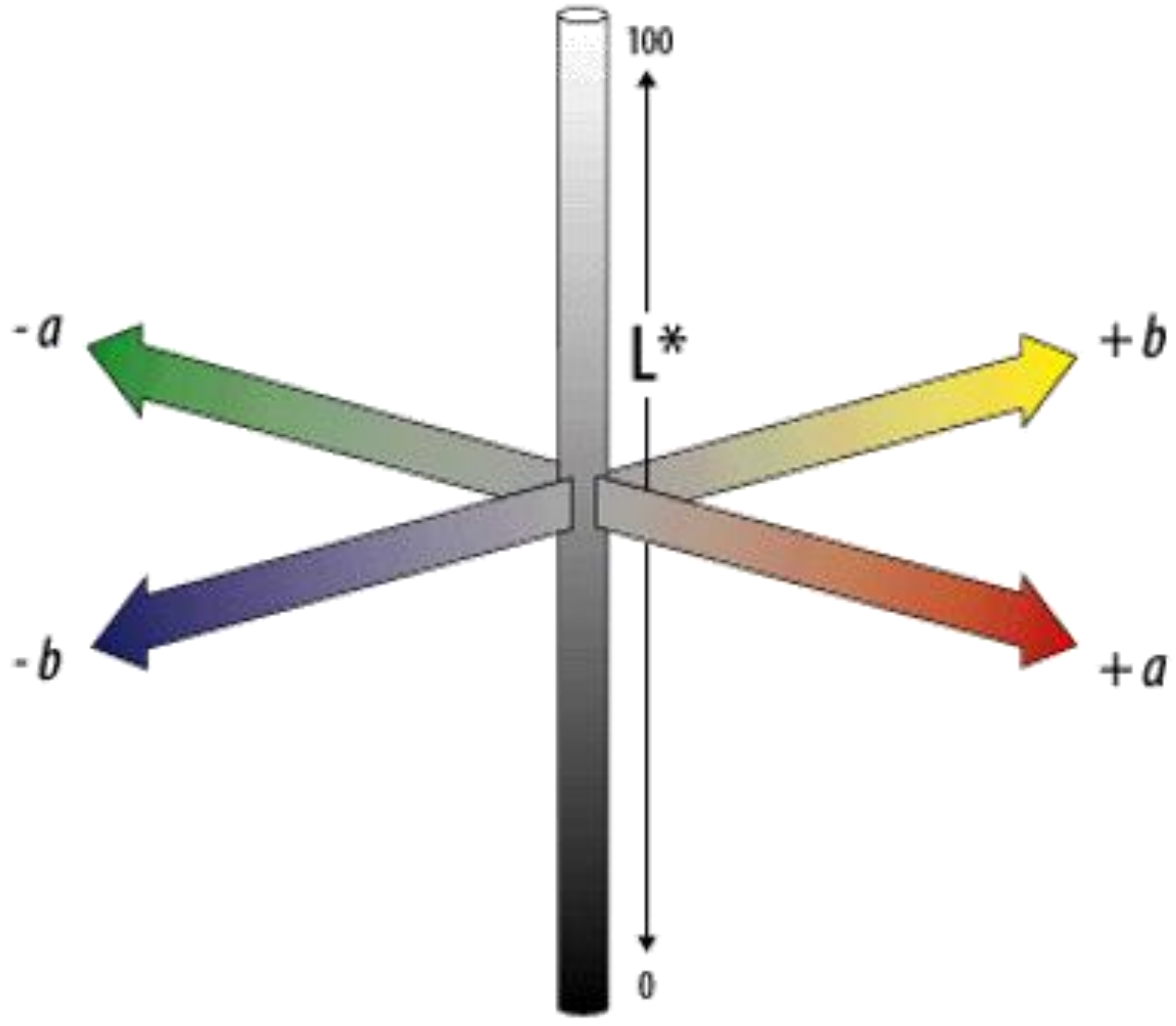
- YPbPr
  - similar to YUV
  - Pb: U scaled
  - Pr: V scaled

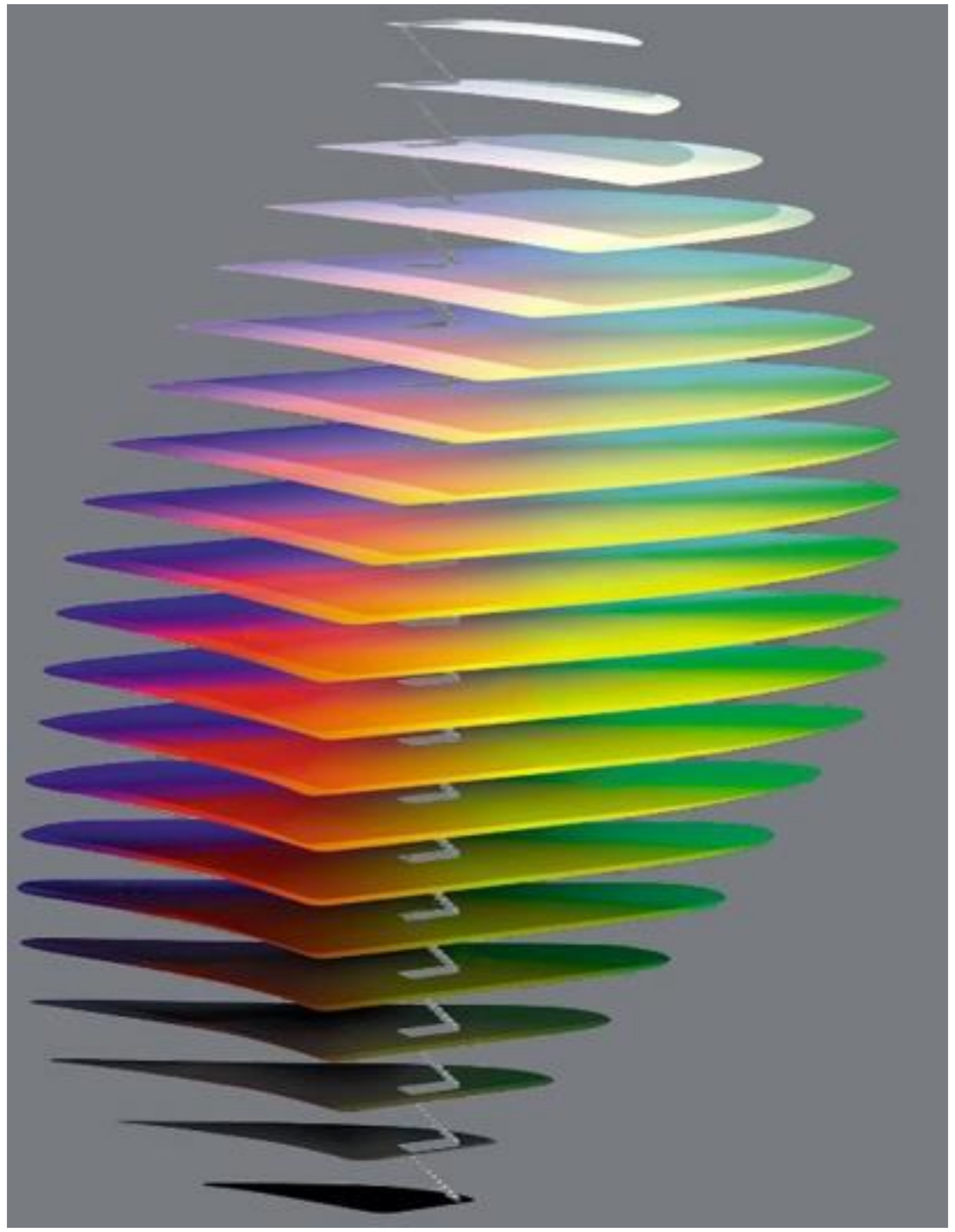
CIE Lab Color Space (1)



- MacAdam's experiment:
  - mark regions of undistinguishable chrominance
  - MacAdam's ellipses
- Apply distortion to transform ellipses to circles

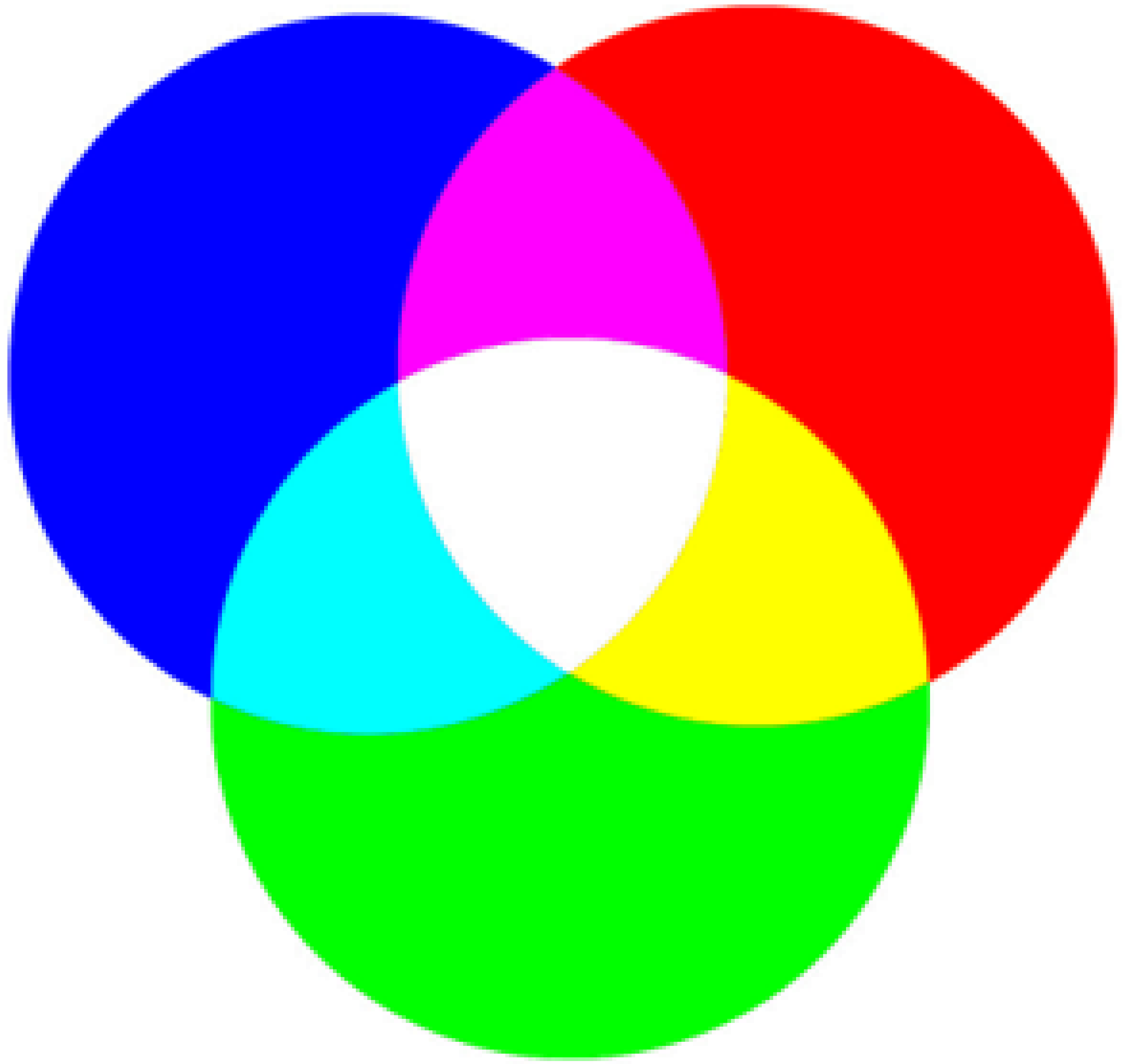
CIE Lab Color Space (2)

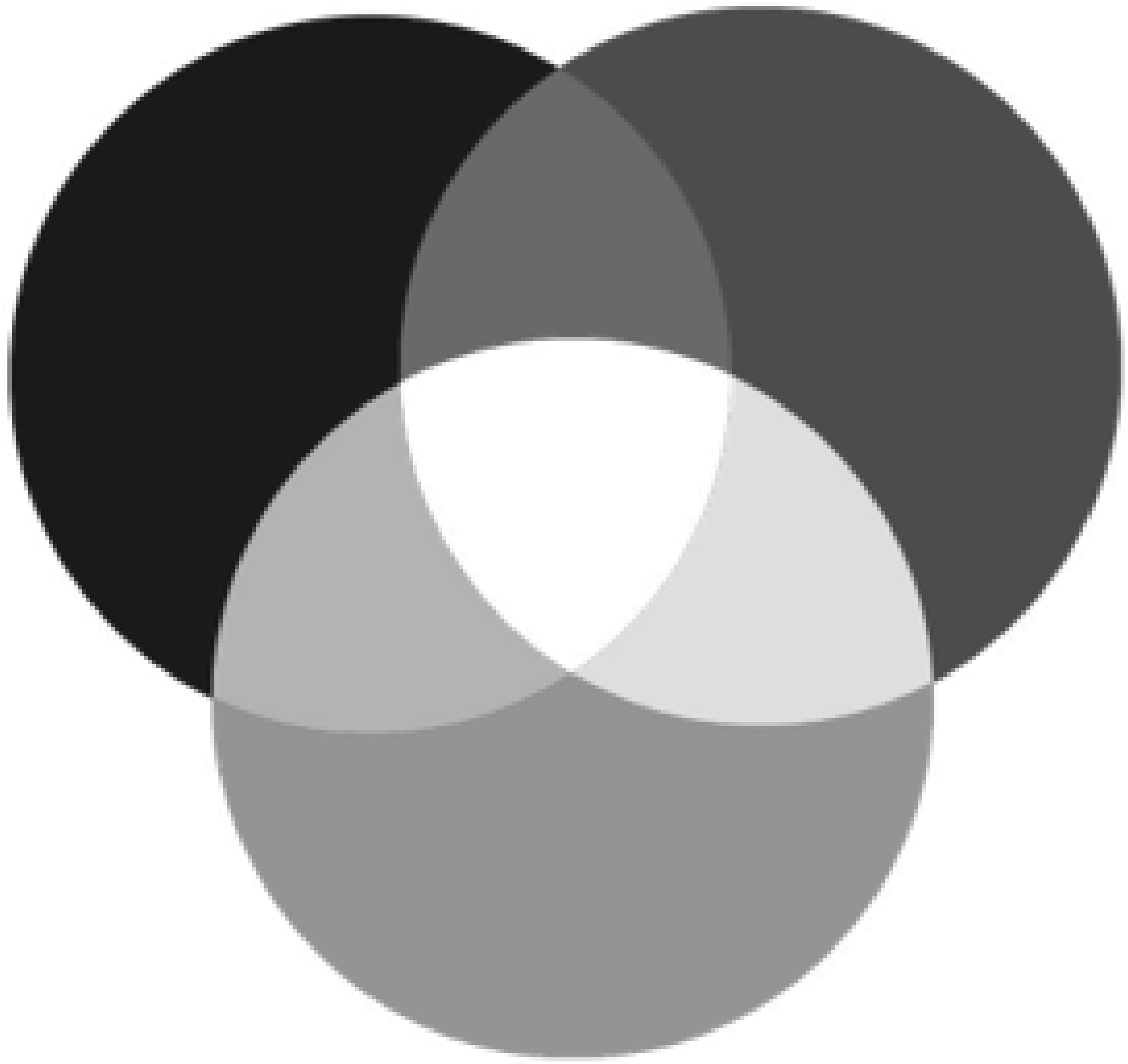




- Attributes
  - L: Luminance
  - a: red-green axis
  - b: yellow-blue axis
- Absolute color system
  numerical changes
  correspond to
  similar perceived changes
- device independent
- widely used
- in the public domain

Color to Greyscale





Compute gray value Y from color values R, G, and B: $Y = 0.299 * R + 0.587 * G + 0.114 * B$

More Information About Colors

- Color Matters
  http://www.colormatters.com
- Educational Color Applets
  http://www.cs.rit.edu/~ncs/color/
- Color glossary:
  http://www.cs.rit.edu/~ncs/color/glossary.htm
- Online book on color
  http://www.colorvoodoo.com/cvoodoo4.html
- Get your own color cube!!!
  http://www.colorcube.com/intro.htm

Exercise 2: Skin Detection

- Segmentation of skin using color information
- Applications:
  - input for face recognition
  - filtering of undesirable images from the internet
- Possible approach
  - use proper color space
  - restrict color range

Exercise 3: Blue Screen

- Replace blue background with different image
- Approach
  - set every pixel's transparency value based on its color
    $a_{ik} = f\ (r_{ik},\ g_{ik},\ b_{ik})$
  - superpose background image with partially transparent image

Exercise 4

Quiz 2 on Ilias